No Country for Honest Men: Balanced Theft, Endogenous Patronage, and the Emergence of Enforcement

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Abstract

A society in which each household steals from another can remain equal when successful raids form a balanced cycle. In *The Black Sheep*, Italo Calvino follows the breakdown of that balance toward hierarchy and policing, and this paper formalizes that sequence. A unit transfer benchmark first shows that balanced reciprocal theft is a weak Nash equilibrium under symmetry. In an infinite-horizon repeated version with stationary Markov behavior, one permanently honest agent destroys stationarity of the original cycle whenever the first night windfall pushes an adjacent household above an exposure threshold. A proportional transfer extension then endogenizes stratification. Rich households prefer delegated theft to personal theft when delegated loot exceeds the worker outside option and contracting cost, and they later replace thieves with guards once concentrated wealth and prey depletion make protection more valuable than further predation. Simulations trace the resulting transition from local asymmetry to class differentiation, patronage, selective protection, rising inequality, and lower utilitarian welfare once theft, contracting, guarding, and punishment consume resources. An isolated honest rupture can therefore dissolve

a reciprocally corrupt order without generating an impartial institutional exit.

Keywords: repeated games, corruption equilibria, patronage, endogenous enforcement, inequality

JEL classification: C73, D74, K42, O17

1 Introduction

In *The Black Sheep*, Calvino begins from a form of order that is plainly criminal and strangely balanced at the same time. Everyone steals at night and everyone is robbed in turn, so each household loses what another household gains and the society remains equal while living under universal predation. The story becomes economically acute when one man decides to remain at home. His abstention removes a link from the nightly circuit and disrupts the accounting on which the circuit rested: the burglar who intended to rob him returns empty handed, while another household, which would otherwise have been robbed by that failed burglar, keeps what it would have lost. The immediate consequence of honesty is therefore a local redistribution. Calvino then follows that redistribution toward a harder order in which some households become rich, poor households are recruited into theft on behalf of the rich, and later the rich hire guards and police to defend fortunes accumulated in the earlier phase (Calvino, 1993). The sequence is literary, but it poses an unusually clean economic problem.

The sequence admits a formal treatment of a regime of reciprocal predation that can reproduce itself as long as participation remains universal, but its initial balance requires exact conditions because a society of thieves remains equal only if every household is both predator and prey under the same accounting. Once one agent abstains, the relevant issue becomes the continuation incentives disturbed by that local break and the path by which reciprocity gives way to a different organization of predation. The later transition raises a further problem, because decentralized theft, delegated theft, and organized protection do not rest on the same primitives. The benchmark that isolates reciprocal balance cannot by itself generate class formation, while the environment that generates class formation obscures the balance from which the sequence begins.

Reciprocal balance is easiest to see in a unit transfer benchmark in which every successful raid

moves one unit, every household begins with the same wealth, and a balanced cycle is a permutation in which every agent has one outgoing raid and one incoming raid. In that environment wealth is preserved period by period, and any such cycle is a weak Nash equilibrium under symmetry. The benchmark remains intentionally narrow because it does not yet contain the ingredients needed for hierarchy or enforcement; its role is to isolate the exact form of reciprocity that Calvino places at the start of the story and to give precise content to the claim that universal participation can sustain equality even inside a predatory order.

The first substantive break comes with the honest perturbation, for which the relevant equilibrium object is a stationary Markov regime in the repeated benchmark rather than a one period game or the full set of repeated outcomes supported by elaborate punishments. One agent is permanently honest and remains at home every period. All other agents choose whether to steal personally or remain at home as a function of current wealth. The first honest night creates a local windfall because one intended victim is no longer robbed, while the predecessor who targeted the honest house fails to obtain loot. Once leaving home exposes wealth left at home, that windfall changes at least one adjacent continuation payoff. Under a transparent wealth threshold condition, the original pattern of balanced reciprocity cannot remain stationary with exactly one permanently honest agent.

Calvino carries the sequence well beyond the first rupture, because a society organized around universal corruption does not move from corruption to honesty in one step but first toward stratified predation and only later toward selective protection. Making that transition economically coherent requires an explanation of why richer households stop stealing personally, why poorer households accept predatory work on behalf of the rich, and why those same elites later prefer guards to thieves. The proportional transfer extension supplies those margins. When theft scales with the wealth of the victim, a rich household that goes out to steal exposes a larger stock at home. Delegation can then dominate personal theft if delegated predation has a targeting advantage and if poor workers face weak outside options. Patron surplus is simply the residual between delegated gross returns and the outside option of the worker net of contracting cost. Later, once elite wealth becomes large and the prey base among the poor is depleted, the value of protection overtakes the value of one more round of

delegated predation. The chronology then moves from reciprocal theft to patronage and from patronage to guarding for reasons that are explicit in the payoff comparisons.

Corruption equilibria remain privately coherent when individual choices change the environment faced by others (Andvig and Moene, 1990; Bardhan, 1997; Shleifer and Vishny, 1993; Tirole, 1996); conventions and equilibrium selection explain how local perturbations move societies across self reproducing orders (Young, 1993); appropriation and protection allocate effort between predation and defense under insecure claims (Skaperdas, 1992; Grossman and Kim, 1995; Hirshleifer, 1995); hierarchy and delegated organization explain how vertical control redistributes rents (Olsen and Torsvik, 1998); and endogenous institutions explain why organized enforcement often protects concentrated advantage (Olson, 1993; North et al., 2009; Besley and Persson, 2009; Acemoglu et al., 2011). The novelty here lies in placing those margins inside one transition from balanced reciprocal theft to patronage and then to selective protection.

The analogy to top trading cycles needs careful handling because it is useful and easy to overstate. A balanced theft cycle resembles a cyclic allocation graph in one limited respect: each agent points outward and is pointed to by exactly one other agent. That graph structure helps explain why the benchmark preserves equality under symmetry and why a local break has consequences beyond the abstainer himself. But the analogy stops at the graph. In the Shapley and Scarf environment, transfers are voluntary and stability is defined through blocking coalitions (Shapley and Scarf, 1974). In the present environment, theft is coercive, generates externalities, and changes later opportunities through exposure and depletion. The cyclic analogy therefore serves only as a disciplined description of structure. Claims about the core or about mechanism design do not survive the change in environment.

Because the benchmark conserves aggregate wealth by construction, aggregate wealth alone cannot distinguish regimes. Period welfare is therefore defined as the sum of concave utilities over wealth minus theft effort costs, contracting costs, guarding costs, and expected punishment losses. This object separates margins that would otherwise be mixed together: pure transfers need not change total wealth, the same transfers can lower utilitarian welfare once they produce inequality, and the later movement from reciprocal theft to patronage and guarding can lower welfare further because predation and pro-

tection consume resources.

The analytical results leave open questions of timing, prevalence, regime duration, and comparative statics that are difficult to settle in closed form once many heterogeneous agents interact, so the same payoff comparisons derived in the theory are embedded in an explicit transition process and studied in simulation. The resulting paths reproduce the first honest perturbation in event time, trace class formation in the proportional environment, map parameter regions associated with patronage, coexistence, guarding, and collapse, and quantify the evolution of inequality and welfare along the transition.

Two restrictions structure the analysis from the outset. No full Markov perfect equilibrium is solved for the large stochastic state space generated by delegated theft, matching frictions, and protection choices,[1] and production is left outside the model. Wealth is redistributed, dissipated through cost, and protected without being endogenously created, which usefully isolates the political economy of predation but also means that the final guard regime should be read as protection of accumulated spoils rather than as a complete productive property-rights order.

The paper proceeds as follows. Section 2 places the argument in the literatures on corruption, conventions, hierarchy, appropriation, and endogenous enforcement. Section 3 presents the unified model, beginning with the unit transfer benchmark and then moving to the repeated perturbation result, the patronage contract problem, the guarding threshold, and the welfare criterion. Section 4 translates those analytical objects into a reproducible simulation design with explicit timing, matching, and regime classification. Section 5 presents the figures and tables in the order required by the argument, from the first honest night to the later regime map and welfare decomposition. Section 6 closes by drawing the discussion and conclusion together before the bibliography and appendix.

2 Related Literature

The problem posed by Calvino belongs to a familiar concern in economics, namely the possibility that conduct which is plainly undesirable in welfare terms can nevertheless form a coherent and durable

[1] This division of labor is deliberate. The theory identifies the margins and thresholds that matter. The simulation studies how those margins interact once the full state space becomes too large for a complete analytical characterization.

equilibrium. In the corruption literature this concern appears in several adjacent forms. Andvig and Moene (1990) show how corruption can reproduce itself by altering the incentives faced by later actors; once others are expected to steal or accept bribes, honest conduct loses much of its private return. Bardhan (1997) surveys the same logic at a broader level and stresses the institutional complementarities that make corrupt systems persistent. Shleifer and Vishny (1993) add the crucial organizational point that the social effects of corruption depend on whether predation is fragmented or coordinated, while Tirole (1996) show how collective reputations can trap organizations in persistent patterns of misconduct even when many participants would prefer a cleaner order. The same concern appears here in a narrower and more sequential setting. Instead of bribery or bureaucratic malfeasance, the object is a world of reciprocal thieves and the transition set off by a small honest rupture that disturbs the accounting on which reciprocity depended.

That emphasis on disturbance and succession also places the analysis near the literature on conventions and equilibrium selection. Young (1993) studies how recurrent interaction and local perturbations can move a society from one sustained convention to another. The relevant insight is that a local change can shift the continuation incentives of neighboring actors and thereby alter the basin of attraction of the entire system. The opening equilibrium in Calvino is a convention in exactly that sense. Its status is behavioral. Normative approval has nothing to do with its persistence. Under symmetry it is regular and reproducible, even though it rests on universal predation. The first honest agent interrupts a pattern of mutual dependence. The first windfall and the first shortfall then alter who still finds participation privately worthwhile. Corruption and equilibrium selection meet at that transition.

The movement from reciprocal predation to hierarchical predation requires a different body of work. A large part of the corruption literature begins with an organization that is already stratified and then studies collusion, monitoring, or internal incentive design. Olsen and Torsvik (1998) remain particularly useful here because they show that hierarchy can alter the distribution of rents and the feasibility of corruption itself. The same problem arises once rich households stop stealing personally and begin to buy predatory labor from poor households. The difference is that hierarchy is not assumed at the start. It has to be generated from the breakdown of a decentralized arrangement. Patronage

therefore requires a strict surplus condition. Delegation cannot rest on vague appeals to elite power alone. It must be profitable because a patron directs theft more effectively than a freelancer can and because the reservation utility of poor workers remains compressed. The central question is how a vertical structure of predation is built out of what had previously been a horizontal order.

Skaperdas (1992) and Grossman and Kim (1995) study environments in which insecure claims induce agents to allocate resources between conflict and defense. Hirshleifer (1995) sharpens the same theme by analyzing the conditions under which anarchy can unravel and more selective forms of order emerge. Demsetz (1967) remains a natural background reference because he frames the emergence of property rights as a response to rising gains from internalization. The connection is close in its treatment of appropriation and protection as alternative uses of resources, but the point of departure matters. The starting point is not a blank condition of disorder. It is an order in which insecurity of claims is universal, reciprocal, and roughly egalitarian. The later rise of guarding is therefore not the first appearance of order. One order replaces another, and the later order protects a distribution created during the transition itself.

The connection to state formation and endogenous enforcement is therefore especially important. Olson (1993) distinguishes roving bandits from stationary ones and argues that an encompassing interest in future extraction can create incentives to provide order. North et al. (2009) generalize that intuition into a theory of violence orders organized around elite management of rents. Besley and Persson (2009) study the conditions under which legal and fiscal capacity emerge, while Acemoglu et al. (2011) show that inefficient coercive states can persist once they serve the interests of those who control them. Acemoglu and Robinson (2008) likewise stress that institutional change often preserves elite power while leaving that power intact. The final regime belongs in that lineage. Guards and police do not arise from civic virtue or from a neutral concern with equal rights. They arise because concentrated wealth becomes worth defending and because labor can be reallocated from predation to protection without changing who controls the terms of social order. The fit with endogenous-institutions work lies there, even though the model remains much more stripped down than the historical theories just cited.

Once guarding enters the picture, Becker (1968) and Becker and Stigler (1974) become relevant because elite households no longer face only the choice between stealing and abstaining. No benevolent state chooses optimal sanctions here, and no optimal policing problem is solved, but the comparison does shift toward two costly technologies, one extractive and one protective, each with its own labor demand and effect on future incentives.

Directed cycles matter here only as a graph theoretic analogy. A balanced theft cycle is also a directed cycle with one incoming and one outgoing link for each agent, and in Shapley and Scarf (1974) directed cycles organize voluntary exchange among agents who reassign indivisible objects. The analogy is useful at that level of structure and no further. The transfers here are coercive, alter later incentives through exposure and depletion, and impose externalities on actors who have not consented to any reassignment, so claims about the core, strategy proofness, or voluntary stability do not survive the change in environment.

Skaperdas (1992) and Grossman and Kim (1995) study how insecure claims divert resources into appropriation and defense. Becker (1968) and Becker and Stigler (1974) keep attention on sanction, precaution, and the incentives of enforcement. Olson (1993), North et al. (2009), Besley and Persson (2009), and Acemoglu et al. (2011) show how organized coercion can stabilize extraction and protect concentrated advantage. Their intersection is the natural place for a setting that begins from egalitarian reciprocal corruption, because aggregate wealth cannot serve as the relevant welfare object once the benchmark preserves total wealth by construction. The comparison has to run through concavity in wealth together with the costs of theft, punishment, contracting, and guarding. Reciprocal corruption, hierarchical predation, and selective protection then appear as successive forms of one order, linked by a local honest rupture that shifts incentives without generating an impartial institutional exit.

3 Model

3.1 Unified architecture and timing

Two layers organize the model. A unit transfer benchmark isolates the balanced accounting at the beginning of the fable by Calvino, with successful theft always transferring one unit, all agents starting symmetric, and the only live actions being to steal or stay home. A proportional transfer extension enters only after symmetry has been broken and adds the margins the benchmark cannot carry on its own: wealth-dependent exposure, contracting between rich patrons and poor thieves, and the later purchase of guards. The unit transfer environment defines the initial cycle and the first honest perturbation; the proportional transfer environment governs stratification, delegated theft, and organized enforcement. Keeping that sequence explicit removes the notational drift that otherwise arises when one environment is asked to carry every result.

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. Agents are indexed by $i \in \{1, \ldots, N\}$ with $N \geq 3$. Wealth at the beginning of period t is $w_{it} \geq 0$. In the benchmark all agents begin with common wealth $\bar{w} > \tau$, where the unit loot is normalized to $\tau = 1$. In the extension the loot from a successful raid on victim j is γw_{jt}, where $\gamma \in (0, 1)$. The notation used throughout the paper is collected in Table 1; the same symbols are used in the theory, the simulations, and the appendix proofs.

Each period begins with observation of the wealth vector $w_t = (w_{1t}, \ldots, w_{Nt})$, continues with action choices, target selection, raids and guard interventions, and ends with the payment of wages, loot shares, contracting costs, guarding costs, and punishment losses that generate the next-period wealth vector w_{t+1}. In the benchmark the action set is $\{\text{steal}, \text{stay}\}$; in the extension, rich agents may additionally choose whether to hire a thief or a guard, and poor agents may accept employment as thieves or guards. This timing fixes the chronology used throughout. The honest agent is protected on the first night because he stays home when the first perturbation occurs. Rich households remain relatively safe in the patronage phase because they are occupied and because dependent thieves have little reason to attack the patrons who employ them. Guards become valuable later because concentrated wealth eventually attracts poaching and because the informal deterrence created by patronage weakens once

Table 1: Notation. The table collects the state variables, contract terms, and regime objects used throughout the theory and simulation sections.

Symbol	Meaning
$i \in \{1, \ldots, N\}$	Agent index.
$t = 0, 1, 2, \ldots$	Time index.
w_{it}	Wealth of agent i at the beginning of period t.
σ	Baseline theft permutation defining the reciprocal cycle.
h	Exogenously honest agent in the perturbation analysis.
τ	Unit loot in the benchmark stage game, normalized to $\tau = 1$.
γ	Proportional theft intensity in the dynamic extension, with $\gamma \in (0, 1)$.
c_T	Resource cost of personal theft effort.
ρ	Exposure coefficient capturing the expected fraction of own wealth lost when one leaves home.
ϕ_t	Targeting advantage of delegated theft relative to freelance theft.
s_{ijt}, α_{ijt}	Fixed payment and loot share in a patron-thief contract between patron i and thief j.
o_{jt}	Outside option of poor agent j if he declines a patron's contract.
k	Contracting cost borne by the patron.
g	Guard wage.
q_t	Probability that guarding blocks a raid on protected wealth.
R_t, E_t, P_t	Sets of rich, elite, and poor agents at date t.

the prey base among the poor is exhausted.

In the benchmark stage game, burglary serves as a reduced form device for balanced reciprocity. The proportional extension restores the margins that matter once wealth diverges. The restricted architecture isolates reciprocity, class formation, and enforcement without introducing production, audit technologies, or a fully solved Markov perfect state space.

3.2 Baseline benchmark: balanced reciprocal theft

Let σ be a permutation of $\{1, \ldots, N\}$ with no fixed points. Under the profile induced by σ, each agent i steals from $\sigma(i)$ and is in turn robbed by $\sigma^{-1}(i)$. The benchmark stage game assumes that a household that leaves home to steal is vulnerable to being robbed, whereas a household that stays home prevents a raid on that date. Since every agent steals in the benchmark profile, every household is absent and every planned raid succeeds. Each successful raid transfers one unit from victim to thief. Wealth at the end of the period therefore satisfies

$$w_{i,t+1} = w_{it} - \tau \mathbf{1}\{\sigma^{-1}(i) \text{ raids } i\} + \tau \mathbf{1}\{i \text{ raids } \sigma(i)\}. \tag{3.1}$$

Along a permutation profile with one incoming and one outgoing raid per agent, both indicators equal one.

Definition 3.1 (Balanced theft cycle)**.** *A balanced theft cycle is a permutation profile σ such that each agent undertakes exactly one outgoing raid and suffers exactly one incoming raid in the period.*

The basic accounting is immediate but worth stating explicitly because it is the benchmark against which everything else will be measured.

Lemma 3.2 (Balanced accounting)**.** *If all agents begin a period with common wealth $\bar{w}$ and play a balanced theft cycle, then every agent ends the period with wealth $\bar{w}$ and aggregate wealth is preserved.*

Proof. See Appendix A.1. Each agent loses exactly τ and gains exactly τ, so wealth is unchanged agent by agent and therefore in aggregate. □

A weak equilibrium notion for the initial regime comes from end-of-period wealth net of the resource cost of personally going out to steal. Let $c_T \geq 0$ denote that cost. In the stripped-down benchmark the cost is small enough that stealing is weakly preferred to staying when all others maintain the cycle. Formally, an agent who follows the cycle ends with $\bar{w} - \tau + \tau - c_T = \bar{w} - c_T$, while an agent who stays home under universal theft avoids being robbed but also forgoes loot, ending with $\bar{w} - \tau$. Thus stealing is weakly better whenever $c_T \leq \tau$.

Proposition 3.3 (Balanced reciprocity as a weak equilibrium)**.** *Suppose all agents begin with common wealth $\bar{w}$, all successful raids transfer $\tau = 1$, and the effort cost of personal theft satisfies $0 \leq c_T \leq \tau$. Then any balanced theft cycle is a weak Nash equilibrium of the one-shot stage game.*

Proof. See Appendix A.1. Under symmetry, deviating to a different victim merely permutes the identity of the counterparty and does not alter the deviator's end-of-period wealth. Deviating to staying home prevents the incoming raid on the deviator's house, but also sacrifices the outgoing loot. The deviation yields $\bar{w} - \tau$, while following the cycle yields $\bar{w} - c_T \geq \bar{w} - \tau$. Hence no unilateral deviation is strictly profitable. □

Only a weak claim is needed at this stage. Universal predation can be mutually consistent and distributively neutral when every agent is both predator and prey in exactly the same way, which is enough to formalize the opening equilibrium in Calvino without claiming efficiency, strictness, or dynamic robustness. For the same reason, this object is not labeled a top-trading-cycle equilibrium. The graph of the benchmark resembles a cyclic allocation problem because every agent has one incoming and one outgoing link, and that resemblance is useful when thinking about local propagation after one link is broken. Yet the resemblance stops at the graph. Here transfers are coerced, victims impose no consent constraint, and one agent action changes the opportunity set of others through exposure and depletion. The benchmark therefore supports only a TTC-like cyclic intuition and does not inherit the incentive and core properties of the Shapley-Scarf environment.

3.3 Honest perturbation and stationary instability

Dynamic analysis requires the horizon and the equilibrium concept to be stated before the result itself. Consider an infinite horizon repeated game with common discount factor $\delta \in (0, 1)$. One agent, denoted h, is exogenously committed to honesty and chooses stay in every period. All other agents can choose either personal theft or staying home in the benchmark environment. Strategies are restricted to be stationary and Markov in current wealth, so each agent conditions only on the current state and not on the full history. This restriction is substantive and is exactly the object needed to address the earlier horizon confusion, because the theorem concerns stationary Markov continuation after the first honest shock and not the full set of repeated equilibria sustained by elaborate punishments.

Let $p = \sigma^{-1}(h)$ be the predecessor who planned to rob the honest agent, and let $q = \sigma(h)$ be the successor whom the honest agent would have robbed under universal participation. Since h stays home at $t = 0$, the raid by p fails, while the raid that would have been undertaken by h never occurs. If every other agent continues the benchmark cycle at $t = 0$, then

$$w_{h1} = \bar{w}, \qquad w_{p1} = \bar{w} - \tau, \qquad w_{q1} = \bar{w} + \tau, \tag{3.2}$$

and all other agents remain at $\bar{w}$.

Lemma 3.4 (First night local asymmetry)**.** *Under the benchmark cycle with one permanently honest agent h, the first honest night creates a local three agent asymmetry given by* (3.2)*. In particular, the successor q receives a windfall and the predecessor p suffers an immediate shortfall.*

Proof. See Appendix A.1. The honest agent neither steals nor is robbed, the predecessor fails in his attempted raid on an occupied house, and the successor is not robbed by the honest agent. □

To determine whether the original regime can continue, the model introduces exposure. If agent i steals personally at date t, he leaves home and faces an expected exposure loss proportional to his own wealth. Let $\rho \in (0, 1)$ denote the exposure coefficient. In the benchmark environment the expected

one-period return from personal theft is

$$\pi^S(w_{it}) = \tau - \rho w_{it} - c_T, \tag{3.3}$$

while the normalized return from staying home is $\pi^N(w_{it}) = 0$. Hence there is a wealth threshold

$$\widehat{w} = \frac{\tau - c_T}{\rho} \tag{3.4}$$

such that an agent weakly prefers personal theft when $w_{it} \leq \widehat{w}$ and strictly prefers staying home when $w_{it} > \widehat{w}$.

Assumption 3.5 (Interior exposure threshold)**.** *The exposure threshold satisfies* $\bar{w} < \widehat{w} \leq \bar{w} + \tau$.

The assumption places the symmetric benchmark wealth level below the exposure cutoff and the first night windfall at or above it. Personal theft is still weakly attractive at $\bar{w}$, but no longer attractive after one unit of windfall. This is not a generic truth about crime. It identifies the narrow parameter region in which the first honest deviation in Calvino has the power to change the best response of the successor immediately.

Theorem 3.6 (No stationary one-honest continuation)**.** *Under Assumption 3.5, there is no stationary Markov equilibrium of the repeated benchmark game that preserves the original balanced theft pattern for all agents other than the permanently honest agent* h.

Proof. See Appendix A.1. By Lemma 3.4, the successor q enters period $t = 1$ with wealth $w_{q1} = \bar{w} + \tau$. Assumption 3.5 implies $w_{q1} > \widehat{w}$, so (3.3) yields $\pi^S(w_{q1}) < 0 = \pi^N(w_{q1})$. Thus q strictly prefers staying home at $t = 1$ and no longer continues to steal. Since the original balanced pattern requires q to keep stealing, that pattern cannot be stationary after the perturbation. □

Under a transparent exposure threshold, the original balanced reciprocity pattern cannot remain a stationary Markov regime once exactly one agent is permanently honest. The theorem does not characterize the full set of repeated equilibria with one honest agent. In the simulation, agents who

inherit the first night windfall stop personal theft once their wealth crosses the threshold implied by (3.4), so the computational rule coincides with the best response stated in the theorem.

Chronology matters here. The honest agent is not robbed immediately upon becoming honest. He is protected on the first night because he stays home, and occupied houses in the benchmark cannot be burgled. Only later, once the society leaves the universal-cycle environment and moves into the stochastic proportional transfer extension, do occupied but unguarded houses become imperfectly safe. The distinction between first night certainty and later stochastic vulnerability is maintained in both the theory and the simulation.

3.4 Proportional-transfer extension and endogenous patronage

Once the first honest shock has broken symmetry, the benchmark no longer suffices. The dynamic extension retains the same population and wealth state, but the theft technology changes. A successful raid on victim j now transfers γw_{jt}, with $\gamma \in (0, 1)$. This is the only theft technology used in the stratification, patronage, and guard regimes. Under unequal wealth, the relevant margin is no longer the gain or loss of one unit. A rich agent who leaves home exposes a large stock of wealth, while a poor agent exposes very little. The proportional specification expresses that asymmetry parsimoniously.

Let m_t denote the expected gross loot from the best personal raid available to a rich agent at date t. For most of the analysis it is enough to treat

$$m_t = \gamma \bar{w}_t^T, \tag{3.5}$$

where $\bar{w}_t^T$ is the expected wealth of the best available target class among nonelite households. If rich agent i raids personally, his expected one-period net payoff is

$$\pi_{it}^{\text{self}} = m_t - \rho w_{it} - c_T. \tag{3.6}$$

If instead he stays home without contracting, the normalized payoff is $\pi_{it}^{\text{stay}} = 0$.

Patronage becomes meaningful only if delegation changes the feasible target set and the division

of surplus. To capture the first margin, let $H_t \subseteq P_t$ denote the set of relatively wealthy but still poorly protected households among the poor. A patron can direct a hired thief toward this set more effectively than a freelancer can. Let $\phi_t \geq 1$ denote that targeting advantage, so delegated gross loot is

$$L_t = \phi_t \gamma \bar{w}_t^H, \tag{3.7}$$

where $\bar{w}_t^H$ is the mean wealth of the targeted high value poor. The model thus keeps the direction of physical theft clear. In the patronage regime, poor thieves usually rob poor or near-poor households, not elite households. The net transfer is poor to rich because some of the loot is remitted upward to the patron.

To capture the second margin, let o_{jt} denote the outside option of poor agent j if he rejects an offer from a patron. The outside option is not the full delegated loot because a freelancer lacks the targeting advantage of the patron and competes with other poor thieves for a depleted prey base. The model writes

$$o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T, \qquad \lambda_t \in (0, 1), \tag{3.8}$$

where $\bar{w}_t^P$ is mean poor wealth and λ_t summarizes congestion and bargaining weakness in the freelance market. A contract between patron i and thief j has fixed payment s_{ijt} and loot share $\alpha_{ijt} \in [0, 1]$ for the thief, while the patron pays contracting cost $k > 0$. The thief accepts if

$$s_{ijt} + \alpha_{ijt} L_t \geq o_{jt}. \tag{3.9}$$

Let $\beta \in [0, 1]$ denote patron bargaining power over the residual surplus. If a contract is formed, total surplus net of the outside option of the thief and the contracting cost of the patron equals

$$\Sigma_t = L_t - k - o_{jt}. \tag{3.10}$$

The payment rule is chosen so that the thief receives his outside option plus the share $(1-\beta)\Sigma_t$, leaving

the patron with $\beta\Sigma_t$. Equivalently,

$$s_{ijt} + \alpha_{ijt} L_t = o_{jt} + (1-\beta)\Sigma_t \tag{3.11}$$

whenever $\Sigma_t \geq 0$.

Lemma 3.7 (Contract feasibility)**.** *There exists a patron-thief contract satisfying the thief's participation constraint if and only if*

$$L_t - o_{jt} > k. \tag{3.12}$$

Proof. See Appendix A.1. A contract is feasible exactly when gross delegated loot exceeds the reservation utility of the thief plus the contracting cost of the patron. □

Condition (3.12) is economically transparent. Patronage is not attractive merely because the patron is rich. It is attractive because delegated predation reaches a target class or success probability that the poor freelancer cannot reach alone, and because the outside option of poor workers is compressed by competition and depletion. If either margin disappears, the contract market collapses.

Proposition 3.8 (Strict patron surplus and occupational choice)**.** *Suppose Lemma 3.7 holds and the patron captures fraction β of the contract surplus. Then patron i's expected payoff from hiring thief j is*

$$\pi_{ijt}^{\text{hire}} = \beta\left(L_t - k - o_{jt}\right). \tag{3.13}$$

The rich agent strictly prefers hiring thief j to both abstention and personal theft whenever

$$\beta\left(L_t - k - o_{jt}\right) > \max\left\{0,\ m_t - \rho w_{it} - c_T\right\}. \tag{3.14}$$

Proof. See Appendix A.1. Equation (3.13) follows directly from the bargaining rule. Hiring dominates abstention because the right-hand side of (3.14) exceeds zero, and it dominates personal theft because the expected surplus from delegation exceeds the net return available to the rich agent who exposes wealth at home. □

Surplus arises because delegation is more productive than freelance theft and because the reservation utility of the worker remains below delegated gross returns. The wedge can come from better targeting, better information, bargaining power, or labour-market slack. The model requires only one such margin to survive.

Poor agents become hired thieves because they remain below the wealth threshold at which exposure makes personal theft unattractive and because the wage or commission implied by (3.11) exceeds their shrinking outside option. The same compression of opportunity that makes them vulnerable to theft also makes them available as predatory labor. That dual role is central to the later emergence of guards.

Corollary 3.9 (Comparative statics of patronage)**.** *Fix w_{it} and suppose $L_t = \phi_t \gamma \bar{w}_t^H$ and $o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T$. Whenever (3.12) holds, patron payoff (3.13) is increasing in ϕ_t, decreasing in λ_t, decreasing in k, and decreasing in the attractiveness of personal theft only through the benchmark on the right-hand side of (3.14). In particular, increases in delegation quality and decreases in poor outside options expand the set of states in which rich agents choose patronage over both self-theft and abstention.*

Proof. See Appendix A.1. The derivative of (3.13) with respect to ϕ_t is positive, the derivatives with respect to λ_t and k are negative, and the comparison against personal theft depends on whether these changes also widen the gap in (3.14). □

These are exactly the margins varied later in the simulation figures. Patronage expands when delegated targeting becomes more effective, contracts when the outside option of poor workers strengthens, and weakens when contracting becomes expensive. The comparative statics in the code therefore match the comparative statics in the model.

3.5 The switch from patronage to guarding

If patronage succeeds for long enough, it changes the environment that made it profitable in the first place. Poor wealth is depleted, so L_t declines over time as the supply of lucrative unguarded targets

shrinks. At the same time, elite wealth rises, making concentrated fortunes worth defending. To capture the latter margin, let $\chi_t \in (0,1)$ denote the residual probability that elite wealth is targeted successfully when it is merely occupied but not formally guarded. This residual risk is low during active patronage because the patron remains at home and because dependent thieves have a reason not to antagonize the patrons who pay them. It need not remain low forever. Once patronage thins the poor prey base and contractual relations unravel, poaching against elites or opportunistic raids by former dependents become more attractive. Occupancy alone no longer replicates the discipline previously created by an active patronage network.

If elite agent i hires a guard at wage g, and guarding blocks a raid with probability q_t, the expected value of protection is the avoided expected loss on concentrated wealth:

$$\pi_{it}^{\text{guard}} = q_t \chi_t \gamma_E w_{it} - g, \tag{3.15}$$

where $\gamma_E \in (0,1)$ is the fraction of elite wealth at stake in a successful elite raid. Poor workers accept guard employment provided the wage weakly exceeds their best alternative, which in the model means

$$g \geq \max\{o_{jt}, \underline{u}_{jt}\}, \tag{3.16}$$

where $\underline{u}_{jt}$ is a noncriminal fallback option. The simulation imposes this acceptance condition directly when guards are hired from the poorest available agents.

The rich agent compares (3.15) with (3.13). Guarding dominates patronage whenever the expected value of avoided elite loss exceeds the surplus from one more round of delegated predation:

$$q_t \chi_t \gamma_E w_{it} - g \geq \beta \left(L_t - k - o_{jt} \right). \tag{3.17}$$

Theorem 3.10 (Endogenous switch to guarding)**.** *Suppose guard labor is available at wage g satis-*

fying (3.16) *and patronage is initially feasible. Then guarding is optimal for elite agent* i *whenever*

$$w_{it} \geq \widehat{w}_t^G \equiv \frac{\beta\left(L_t - k - o_{jt}\right) + g}{q_t \chi_t \gamma_E}. \tag{3.18}$$

Moreover, $\widehat{w}_t^G$ *falls as the prey base among the poor is depleted, because* $L_t - o_{jt}$ *declines with* $\bar{w}_t^H$ *and* $\bar{w}_t^P$*.*

Proof. See Appendix A.1. Rearranging (3.17) yields the threshold. Depletion reduces the right-hand side of (3.17), lowering the elite wealth level at which guarding becomes privately optimal. □

Several timing and role ambiguities also disappear under this threshold comparison. Rich households need not hire guards at the outset because elite wealth is not yet concentrated enough and active patronage already creates informal deterrence. Guards emerge later even though rich households were occupied earlier because occupancy no longer suffices once concentrated wealth becomes a tempting target and patronage ceases to discipline the population that once supplied predatory labor. The direction of predation also remains clear: during patronage, poor thieves are usually robbing poor households for rich principals, and once guarding dominates the same pool of poor agents may be reallocated from predation to protection.

Corollary 3.11 (Comparative statics of guarding)**.** *The guarding threshold* $\widehat{w}_t^G$ *in* (3.18) *is decreasing in guard effectiveness* q_t*, decreasing in elite exposure* χ_t *and elite raid severity* γ_E*, increasing in the guard wage* g*, and increasing in the residual surplus from patronage* $\beta(L_t - k - o_{jt})$*. Guarding therefore becomes more likely when elite wealth is highly exposed and easily defended, and less likely when patronage remains highly profitable or guard labor becomes expensive.*

Proof. See Appendix A.1. The claims follow from direct differentiation of (3.18) with respect to the relevant parameters. □

Stronger guarding technology or greater elite exposure expands the region in which protection is privately optimal. Stronger patronage surplus or more expensive guard labor contracts that region. The move to policing is therefore an endogenous choice over technologies of control.

3.6 Welfare

The welfare criterion has to capture inequality and resource dissipation at the same time. An incomplete honest shock can move a society organized around reciprocal corruption toward more unequal and more resource-intensive regimes, while aggregate wealth alone remains uninformative because the benchmark conserves total wealth by construction. The analysis therefore uses discounted utilitarian welfare with explicit resource costs:

$$W_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t \mathcal{W}_t, \qquad \mathcal{W}_t = \sum_{i=1}^{N} u(w_{it}) - c_T n_t^S - k n_t^H - g n_t^G - \Psi_t, \tag{3.19}$$

where u is strictly increasing and strictly concave, n_t^S is the number of personal thieves, n_t^H is the number of active patron-thief contracts, n_t^G is the number of guards, and Ψ_t denotes expected punishment losses.

Proposition 3.12 (Redistribution, inequality, and resource cost)**.** *Suppose aggregate wealth $\sum_i w_{it}$ is fixed at date t and u is concave. Then any mean preserving spread of the wealth distribution weakly lowers $\sum_i u(w_{it})$. If, in addition, two regimes with the same aggregate wealth differ only in that one has weakly greater inequality and strictly greater values of n_t^S, n_t^H, n_t^G, or Ψ_t, then the latter regime yields strictly lower period welfare $\mathcal{W}_t$.*

Proof. See Appendix A.1. The first claim is Jensen's inequality. The second follows because the cost terms enter linearly and negatively in (3.19). □

A balanced corruption regime can already be normatively poor if theft effort and punishment risk consume resources, even when the distribution remains equal. Once the honest perturbation generates inequality and induces contracting and guarding costs, welfare can fall even if total wealth is approximately conserved in the short run. The relevant object is therefore distribution plus risk plus deadweight cost, and not redistribution alone.

Corollary 3.13 (Guarding may improve welfare relative to patronage)**.** *Fix the wealth distribution at date t. If adopting guarding lowers expected theft effort, contracting activity, and punishment*

losses by more than the resulting guard wage bill, then guarding raises $\mathcal{W}_t$ relative to patronage even though it need not restore the welfare of the original balanced benchmark.

Proof. See Appendix A.1. The difference in period welfare is the reduction in theft-related costs minus the new guarding expenditure. □

Organized enforcement can be privately attractive for elites and even socially preferable to active patronage on a utilitarian measure, while still leaving society worse off than the initial equal benchmark once one accounts for the path by which elite wealth was accumulated. The model therefore leaves room for a limited welfare improvement from policing relative to patronage without mistaking that improvement for a return to egalitarian order.

A balanced theft cycle is a coherent benchmark in the unit transfer stage game. A single permanent honest perturbation breaks that benchmark in the repeated game when wealth-dependent exposure is strong enough. After symmetry breaks, proportional theft together with weak poor outside options makes patronage feasible and profitable. Later, depletion of the prey base and concentration of elite wealth make guarding privately optimal. The analytical results identify those margins, and the simulations trace their timing, prevalence, and welfare consequences.

4 Simulation Design

The analytical model identifies the margins that matter, but it does not determine a unique stochastic path once many heterogeneous agents interact. After the first honest night, wealth dispersion changes occupational choices, occupational choices change target vulnerability, patronage and guarding reshape the pool of available workers, and the resulting history feeds back into later wealth. A closed form characterization of all continuation paths would require a large stochastic dynamic game with endogenous role assignment. The simulations translate the analytical objects from Section 3 into an explicit transition process and then measure the timing, prevalence, and comparative statics of the resulting regimes.

4.1 State variables, timing, and implementation

The simulation state in period t contains four elements: the wealth vector w_t, the role vector r_t, the contract network among patrons and thieves, and the guard network among elites and guards. Roles take values in $\{$honest, stay, freelance thief, hired thief, patron, guard, elite$\}$, with the understanding that an elite is also a patron or protected household when the relevant threshold is met. The state is updated according to the same period timing used in the theory. Wealth is observed first. Rich and elite sets are then classified by threshold rules that use current mean wealth. Rich agents compare the payoffs from abstention, personal theft, and patronage. Elite agents compare patronage to guarding using the threshold in (3.18). Poor agents decide whether to accept theft or guard employment according to their participation constraints. Targets are then selected, raids occur, payments are realized, and the next-period state is recorded.

Deterministic code handles the exact event accounting required by the first honest night, while stochastic code governs the richer transition once matching, targeting, and enforcement interact. The baseline seed is fixed at 613, every sweep uses reproducible seed offsets tied to the parameter grid, and every displayed result is generated from the same underlying pipeline without hand-edited intermediates.

By aligning all paths around the first honest night, event time separates the initial perturbation from later adaptation. Many transition models report only long run outcomes, which makes it difficult to distinguish the direct consequence of an initial perturbation from the later consequences of endogenous adjustment. Here the first honest night is isolated in its own deterministic routine, and the later stochastic dynamics are aligned relative to that event. Parameter changes can therefore be read through their channel: amplification of the first local asymmetry, a change in the profitability of contracting, or a change in the private value of protection.

4.2 Mapping the theory into simulation rules

To keep the logic of Theorem 3.6 fully visible in event time, the benchmark simulation stays very close to the theorem itself. Agents begin on a balanced cycle in the unit transfer environment, one

agent is permanently honest, and on the first night the honest agent stays home while the original cycle is otherwise followed exactly. This reproduces the accounting in Lemma 3.4. In later periods, agents compare their wealth to the threshold in (3.4). If wealth lies below the threshold, they continue personal theft along the cycle; if wealth exceeds the threshold, they stay home. No contracting, guarding, or stochastic targeting is yet present, so any transition away from the balanced cycle can be attributed directly to the honest perturbation.

Once proportional transfers are introduced, role thresholds are defined relative to current mean wealth. An agent becomes rich when his wealth exceeds $\theta_R \bar{w}_t$, where $\bar{w}_t$ is the cross-sectional mean, and becomes elite when wealth exceeds the higher threshold $\theta_E \bar{w}_t$. These thresholds are reduced form devices that translate the payoff comparisons of Section 3 into a finite-agent simulation. Rich agents are those for whom the exposure term ρw_{it} makes personal theft relatively unattractive, while elites are those whose stock of wealth makes guarding salient. In the baseline calibration, the thresholds preserve event time separation between the first honest shock and later institution building while still allowing the patronage regime to emerge within the simulation horizon.

Rich agents are sorted by wealth, and candidate thieves are drawn from the poorest available agents who are neither permanently honest nor already employed. A candidate contract is formed only if the feasibility condition in Lemma 3.7 holds with current state variables. When a contract is feasible, the simulation applies the bargaining rule from (3.11): the thief receives his outside option plus the share of the residual surplus not captured by the patron, and the patron receives the rest. The thief then targets households in the high value poor set H_t, while freelancers target vulnerable households by expected loot. This distinction matters because it is what turns patronage into a strict improvement over freelancing for the patron. Delegated theft is more selective, and its returns are split through a contract that reflects the poor worker outside option.

The targeting rule also resolves a confusion that often arises in informal retellings of the story by Calvino. In the patronage regime, poor thieves typically rob poor households for rich patrons. That is, the physical direction of theft is still largely horizontal among the nonelite, but the financial direction of net transfer is vertical because part of the loot is remitted upward. The simulation keeps this distinction

explicit by defining a candidate target set that excludes the patron himself and other currently protected elite households. The house of the patron is occupied and therefore hard to raid, and a hired thief who turns immediately on the employer would be leaving the intended regime. Such betrayals do occur later through the residual elite-risk term χ_t, but they are not the normal action in the patronage phase.

Elites then face a second occupational comparison, namely whether the expected surplus from another patron-thief contract is smaller than the expected value of avoided elite loss from a guard. If inequality has risen and the prey base among the poor has thinned, the guarding condition in (3.18) will be satisfied for some elite households. Guards are again recruited from the poorest available agents, but only if the guard wage exceeds their best alternative. Once hired, a guard reduces the success probability of raids on the protected house and introduces a punishment probability if an attempted elite raid fails. This punishment channel matters for two reasons. It creates a cost wedge between attacking guarded and unguarded households, and it gives the simulation a concrete mechanism by which formal protection suppresses open theft without eliminating the underlying population of agents willing to steal.

Across the simulation, chronology runs through three stages that match the theory. The first stage is the deterministic first night perturbation in the unit transfer benchmark. The second is the patronage stage in the proportional transfer environment, where class differentiation and delegated predation emerge. The third is the enforcement stage, where some elites replace delegated theft with guards and punishment. Endogenous threshold crossings hard-code the chronology, so calendar dates do not drive the switch and the timing of the transition becomes an outcome determined by wealth dispersion, target depletion, outside options, and enforcement effectiveness.

4.3 Baseline calibration

The baseline calibration aims at plausibility within the internal units of the model and is not empirically estimated from a specific dataset. The number of agents is set to $N = 60$, which is large enough to produce meaningful cross-sectional dynamics and small enough to make network visualizations interpretable. Initial wealth is normalized to one for every agent. The proportional theft intensity is

Table 2: Baseline simulation parameters. The values define the benchmark environment used in the main figures, while the parameter sweeps vary the key margins systematically around this baseline.

Symbol	Value	Description
N	60	Number of agents
γ	0.350	Proportional theft intensity
$\bar{w}$	1	Initial wealth per agent
θ_R	1.150	Rich threshold multiplier
θ_E	2.100	Elite threshold multiplier
s	0.030	Base contract wage
α	0.450	Thief loot share under contract
κ	0.030	Contracting cost
g	0.080	Guard wage
q	0.980	Guard effectiveness

$\gamma = 0.35$, so a successful raid extracts a substantial but not total fraction of the victim wealth. The rich threshold is $\theta_R = 1.15$ times current mean wealth, while the elite threshold is $\theta_E = 2.10$ times current mean wealth. These values imply that early windfalls matter but do not instantly produce an elite stratum.

The remaining parameters pin down the costs and comparative advantages of the occupational choices. The cost of theft effort is $c_T = 0.02$. The contracting cost is $k = 0.03$. The baseline guard wage is $g = 0.08$. The target thief share under a contract is $\alpha = 0.45$, while the fixed wage component is calibrated at $s = 0.03$. The success probability of raiding an occupied but unguarded household is low but positive, and the success probability against a guarded elite household is lower still. Punishment after a failed guarded raid is likely but not certain. In words, the calibration assumes a society in which unprotected wealth is vulnerable, protection is costly, and the poor can be induced to work as thieves or guards because their outside options are weak. Table 2 reports the core baseline values used in the main figures.

Parameters that govern the first night logic are chosen to mirror the benchmark theory rather than to generate dramatic inequality mechanically. Parameters that govern patronage and enforcement are chosen so that each regime can emerge for reasons consistent with the propositions in Section 3. For example, the baseline outside option of poor thieves is set low enough that patronage can be strictly profitable, but not so low that every rich agent hires a thief immediately. Likewise, guard effectiveness

is set high enough that protection can dominate when wealth is sufficiently concentrated, but not so high that guards appear before a meaningful patronage phase has existed.

The baseline path is kept moderate: inequality rises without exploding instantly, guards appear only after a visible patronage phase, and collapse remains possible without becoming inevitable. A more aggressive calibration would produce sharper pictures, but it would also obscure whether the mechanisms emerge from the stated thresholds or from extreme parameter values.

4.4 Parameter sweeps and regime classification

Systematic parameter sweeps then vary the margins highlighted in the theory. Theft intensity γ is varied to measure how sensitive class formation is to the fraction of wealth that can be appropriated in a single raid. Exposure risk is varied through the success probability of attacks on occupied but unguarded households, which determines how costly it is for a rich household to rely on occupancy alone. Contracting cost k and patron bargaining power β are varied to map the feasibility and profitability of delegation. Guard effectiveness and punishment severity are varied to locate the transition from patronage to protection. The outside option of poor thieves is varied through the congestion parameter λ_t and the availability of guard employment. Finally, the size of the initial honest set is varied so that the simulations can distinguish a single local perturbation from a broader coordinated honesty shock.

Each period in each simulation path is classified into one of six regimes: balanced cycle, transition, patronage, guard regime, coexistence, or collapse. The balanced-cycle label is reserved for periods in which almost all agents continue the reciprocal pattern and inequality remains negligible. Transition denotes periods after the first honest disturbance but before patronage or guarding has become dominant. Patronage denotes periods with active patron-thief contracts and no meaningful guard presence. Guard regime denotes periods in which protected elites and guards are established at scale. Coexistence denotes periods in which patronage and guarding coexist in nontrivial numbers. Collapse denotes periods in which the poor prey base is so depleted that predatory and protective roles no longer reproduce the earlier structure. These labels are not rhetorical. They are deterministic functions of role counts, inequality thresholds, and guard presence stored in the simulation output. The phase diagram

in Section 5 is built directly from these classifications.

Fixed coding rules determine regime classification. Regimes are not inferred from unsupervised clustering, and paths are not relabeled ex post to fit the figures. The same classification rule is imposed across the entire parameter grid, which is necessary if a phase diagram is to carry comparative meaning.

4.5 Recorded outcomes

For each path the code stores agent level wealth, role histories, employer-worker links, target links, guard assignments, realized and expected welfare components, inequality measures, and first entry times into patronage and guard states. This makes it possible to construct several types of figures that the theory alone cannot provide: event time wealth trajectories by eventual role, distributional heatmaps and quantile ribbons, Lorenz curves at key dates, phase diagrams across parameter pairs, and network snapshots of theft, patronage, and guarding. The same stored objects support the tabulated summaries of first patron entry, first guard entry, final inequality, and dominant long run regime.

Recorded outcomes answer a limited set of analytical questions tied to the theory: how the first honest perturbation propagates through the wealth distribution, when delegated predation becomes organizationally stable, and when organized protection replaces or disciplines predation. Event time plots establish the timeline. Distributional plots establish stratification. Comparative statics show that patronage and guarding do not depend on one baseline path. Welfare decompositions track what is lost and what, at times, is partially recovered as regimes change.

5 Simulation Results

5.1 The first honest night

Balanced reciprocity first breaks through an exact accounting, which is why Figures 1 and 2 precede any discussion of long horizon inequality or institutional emergence. Figure 1 follows the honest agent, the predecessor, the successor, and the remaining population in event time. The structure is exactly the one derived in Lemma 3.4. The honest agent does not suffer an immediate loss because he is at home

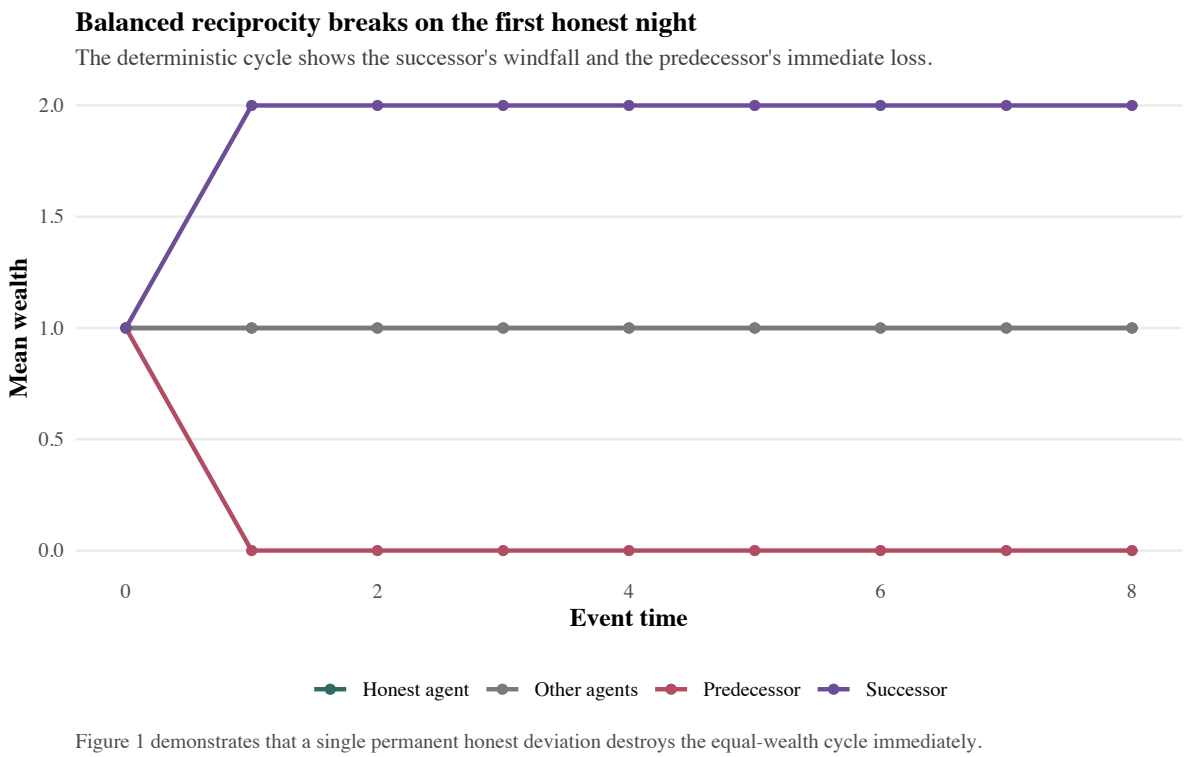

Figure 1: Balanced reciprocity breaks on the first honest night. The successor of the honest agent receives an immediate windfall, while the predecessor who attempts to rob the honest house returns empty handed. The figure demonstrates the local asymmetry that starts the transition.

on the first night. The predecessor who attempts to rob that occupied house receives no loot and is still robbed by his own predecessor, so his wealth falls. The successor, who would otherwise have been robbed by the honest agent, keeps the proceeds of his own raid and receives the first windfall. The first break therefore takes the form of a precise redistribution over a three agent neighborhood of the original cycle.

Figure 2 compresses the same event into a cleaner accounting display. The honest agent is protected on the first night. The first winner is the successor, not the abstainer. The first loser is the predecessor who raids an occupied house and returns with nothing. That chronology matters because the stationary instability theorem rests on the successor windfall. The state change pushes at least one neighboring agent above the exposure threshold. Immediate victimization of the honest agent would produce a different theorem and a different simulation path.

Because the target structure contains no noise, the departure from the original cycle can be attributed

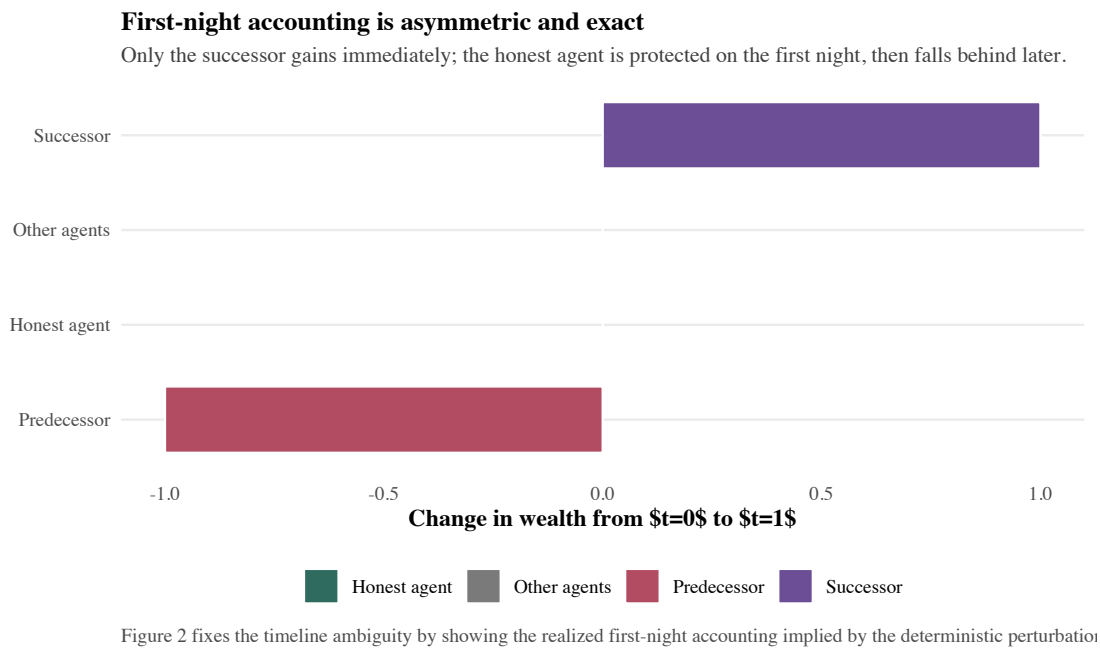

Figure 2: First night accounting. The honest agent is protected on the first night and is robbed only later in the transition; the winners and losers on that night are therefore the successor and predecessor in the original cycle. The figure demonstrates the exact timeline that underlies the instability theorem.

to equilibrium logic instead of chance. Once the successor wealth exceeds the threshold in (3.4), the continuation path begins to involve further choices to stay home and new interruptions in the network. That is exactly the mechanism stated in Theorem 3.6, and the event time display allows the reader to watch the theorem unfold in visible form.

5.2 From local asymmetry to class formation

The longer horizon simulations show that the first honest shock does not merely reshuffle labels on who is temporarily ahead and who is behind. It generates persistent class structure. Figure 3 groups agents by the highest institutional role they reach during the transition and then traces the resulting event time wealth paths. That choice of grouping matters because many agents later withdraw into staying or inactivity, which would otherwise hide the earlier separation. The split appears early. Agents who ever become patrons pull away almost immediately after the initial disturbance, while agents who later serve as guards remain well below them and the honest household falls away sharply. Differentiation therefore begins before the late path settles into its quieter occupational pattern. That is consistent with

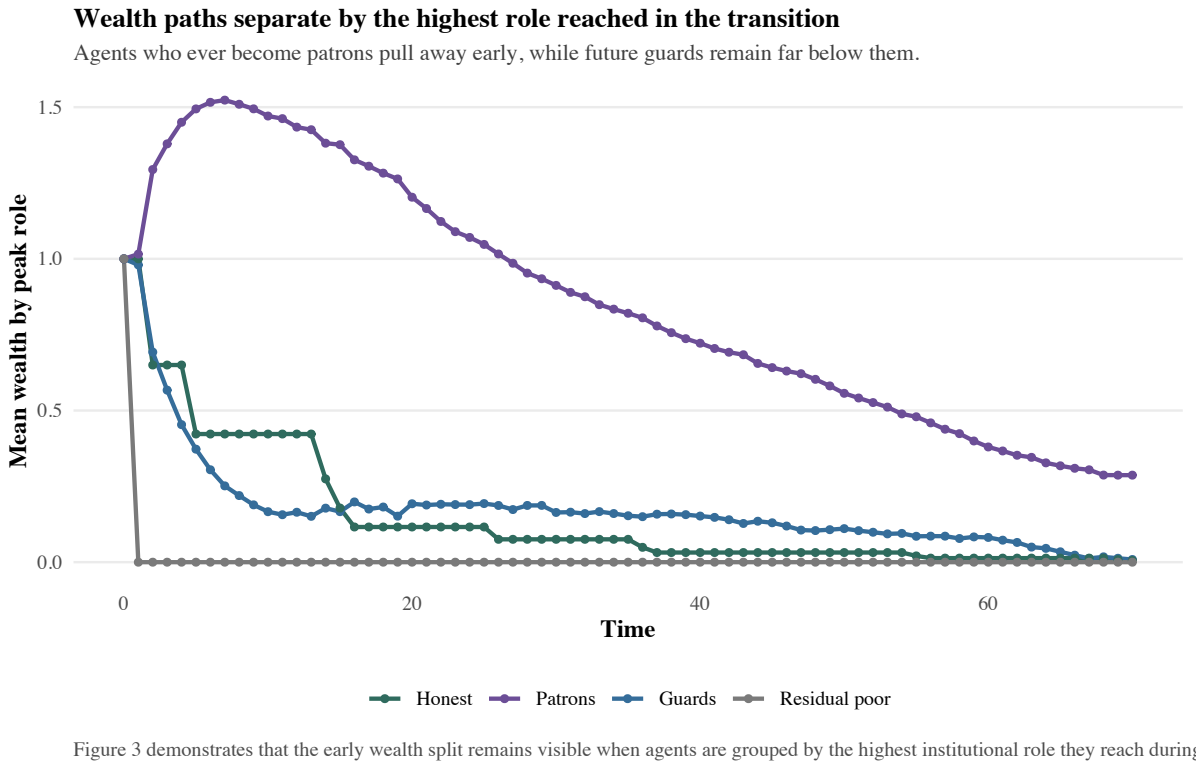

Figure 3: Event time wealth trajectories by peak role in the transition. Agents who ever become patrons separate quickly from future guards and the residual poor. The figure demonstrates that class differentiation appears early in the transition and is not an artifact of the terminal date.

the theory. Rich households emerge first because personal exposure changes the occupational choice of early windfall recipients. Patronage enters after that distribution has already begun to separate.

Figure 4 shows the same process in cross section. The heatmap and quantile ribbons reveal a distribution that rapidly loses its middle. In early periods the support remains narrow because the honest perturbation is still local. A few periods later the upper tail thickens while the lower quantiles bend downward as repeated raids and weak outside options trap a growing mass of agents in the poor set. The widening is not symmetric. The upper tail is carried by a small number of agents who stop exposing their own wealth and later capture part of delegated loot, while the lower tail contains agents who continue to circulate through risky occupations and increasingly depleted target pools. The simulation is therefore producing more than heterogeneity. It is producing a different allocation of risk, power, and occupational choice across the distribution.

Figure 5 translates that cross sectional movement into conventional distributional statistics. The

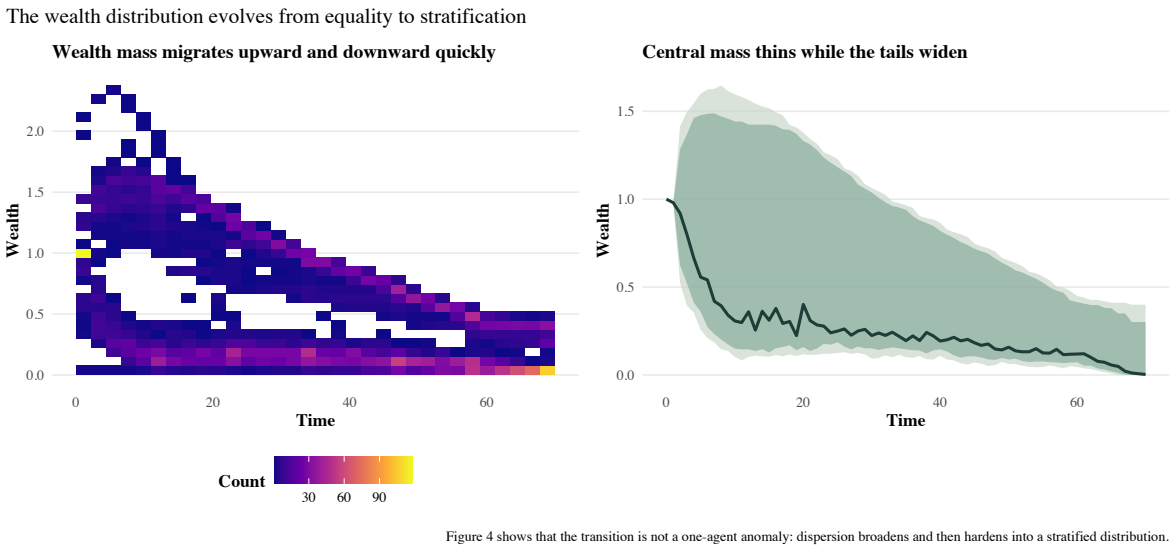

Figure 4: Evolution of the wealth distribution. The heatmap and quantile bands show the rapid thinning of the middle and the thickening of both tails. The figure demonstrates that the honest perturbation produces durable stratification instead of a temporary reshuffle.

Gini coefficient rises sharply from its benchmark value near zero. The top decile wealth share increases in tandem. Lorenz curves at selected dates move steadily away from the forty five degree line. The summary statistics matter because they show that the transition is not confined to a few pathwise anecdotes. The distributional consequences are large, systematic, and persistent in the baseline calibration.

What these displays rule out is a simple narrative in which the move from honesty to inequality is merely literary convenience. In the model, the deterministic perturbation creates a state difference, the exposure threshold converts that difference into differentiated behavior, and the proportional environment amplifies the behavioral difference because the rich internalize exposure while the poor remain available for predatory labor. The computational output supports the theoretical sequence and keeps it from collapsing into a black box path.

5.3 Patronage and the surplus from delegation

Across an interpretable range of parameters, rich agents hire poor thieves only when the contract creates surplus that is both feasible and privately valuable. Figure 6 examines that condition directly. The left panel varies the targeting advantage of delegation. The right panel traces the gap between compensation

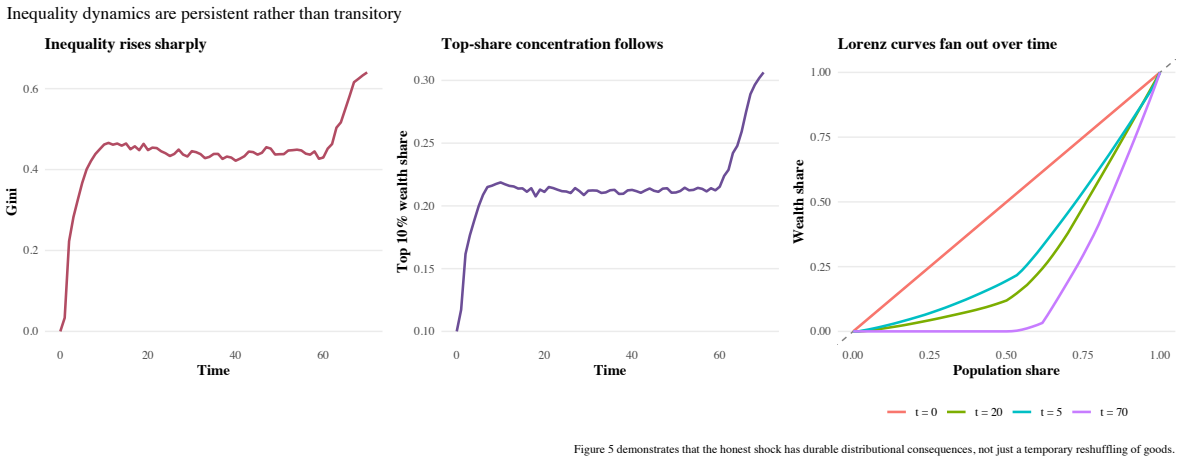

Figure 5: Inequality dynamics. The Gini coefficient, top decile wealth share, and Lorenz curves all move sharply away from the egalitarian benchmark. The figure demonstrates that the transition has strong and persistent distributional consequences.

and the outside option as worker slackness changes. Patron surplus rises with the first margin and falls as the second margin strengthens, exactly as Proposition 3.8 predicts. Delegated theft that is better targeted than freelancing leaves room to satisfy participation and still retain a positive residual. A stronger outside option narrows that residual and can eliminate it.

Delegation becomes profitable only because the environment generates a clear wedge between delegated returns and the outside option of the thief. If the thief could always obtain alone the same gross loot he could obtain for the patron, and if the patron had neither targeting advantage nor bargaining power, patronage would collapse to a zero-profit transfer. Here the patron directs predation more effectively or protects the contract well enough to make delegation more productive than freelancing, while the thief operates in a crowded and depleted market. The gap between delegated returns and the outside option pays the contracting cost and still leaves a residual for the patron.

Patronage need not be monotone in every parameter that raises expected loot. When theft intensity becomes too large, the prey base can be depleted quickly, which shortens the durability of the contract regime even when delegation is initially attractive. Proposition 3.8 gives the static dominance condition; the simulations trace its interaction with depletion, competition, and regime duration.

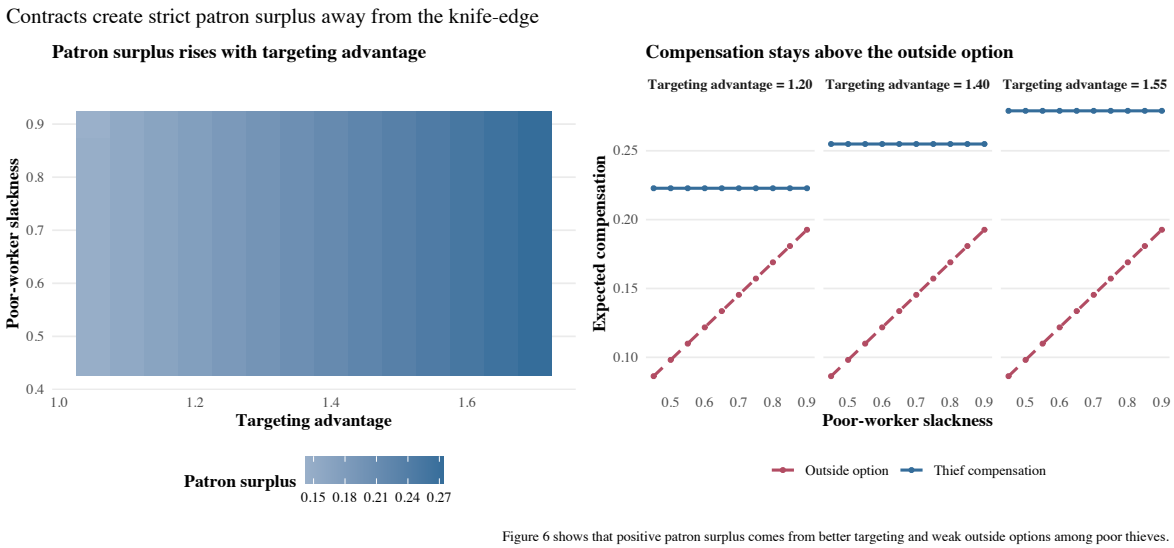

Figure 6: Contracting comparative statics. Patron surplus rises when delegated theft has a stronger targeting advantage and poor outside options are weaker. The figure demonstrates the explicit source of patron surplus and removes the earlier zero profit knife edge.

5.4 From patronage to guarding

Figure 7 maps the dominant medium run regime over the parameter plane spanned by theft intensity and enforcement effectiveness. Several patterns are worth emphasizing. Low theft intensity combined with weak enforcement tends to preserve a long transition because the returns to both predation and protection remain limited. As theft intensity rises and enforcement becomes more effective, the dominant regime shifts toward coexistence and guarding. Collapse appears where predation is intense enough and protection weak enough that the social base is depleted before a stable protected elite can consolidate. This is the computational analogue of Theorem 3.10. Once concentrated wealth becomes valuable enough to protect and protection becomes sufficiently effective, elites prefer to preserve their claims and stop financing one more round of delegated extraction.

On this particular parameter plane patronage appears primarily as a transitional corridor and only infrequently as the dominant medium run regime. The diagram is therefore populated mainly by transition, coexistence, guard, and collapse regions. Patronage remains visible in the event time paths, the network snapshots, and the contracting comparative statics. The phase diagram adds the further point that patronage need not survive as the dominant institutional form once enforcement becomes effective

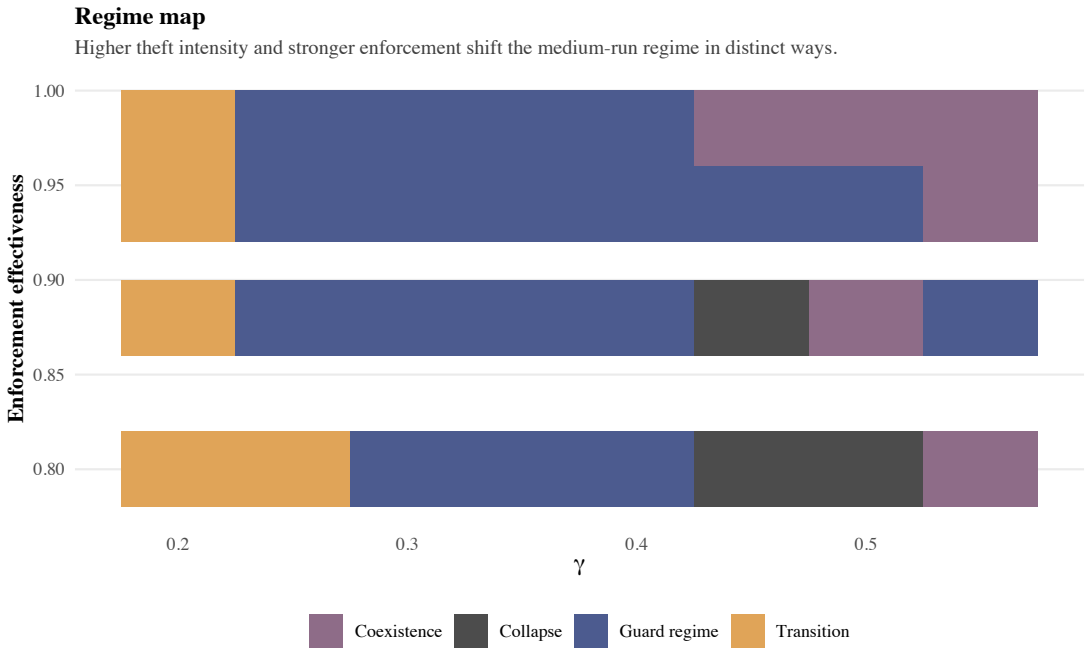

Figure 7: Regime phase diagram. Different combinations of theft intensity and enforcement effectiveness lead to distinct medium run transition, coexistence, guard, and collapse regions. The figure demonstrates that the successor regimes of the model occupy interpretable parts of the parameter space and do not arise from a single baseline path.

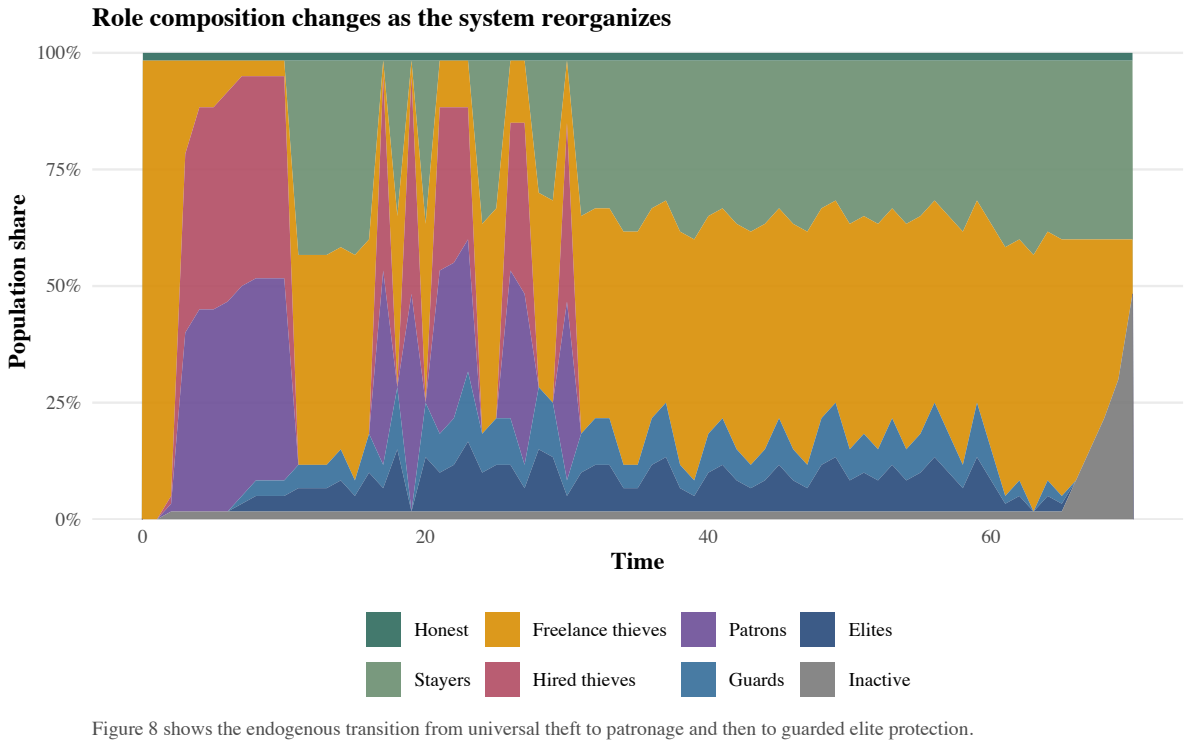

Figure 8: Role shares over time. Universal theft gives way first to patronage and later to selective protection, with guarding remaining present through much of the enforcement phase. The figure demonstrates the endogenous reorganization of the social hierarchy after the first honest shock.

enough or depletion becomes severe enough.

Figure 8 restores time to the same regime logic. The share of agents engaged in direct or freelance theft falls after the first perturbation. Patrons and hired thieves rise next. Guards appear later and remain active through much of the enforcement phase, even though their share is not monotone over time. The plot works as an occupational decomposition of the whole mechanism. The early decline in universal theft reflects the exposure logic of Theorem 3.6. The rise in patron and hired thief roles reflects the surplus logic of Proposition 3.8. The later appearance of guarding reflects the threshold logic of Theorem 3.10.

Figure 9 separates the part of the transition that is pinned by chronology from the part that is not. Under the simulation timing, the first opportunity to hire arrives immediately after the deterministic honest shock, and every run generates at least one patron at that first post-shock contracting round. A raw first-entry statistic for patronage is therefore uninformative. The left panel instead measures

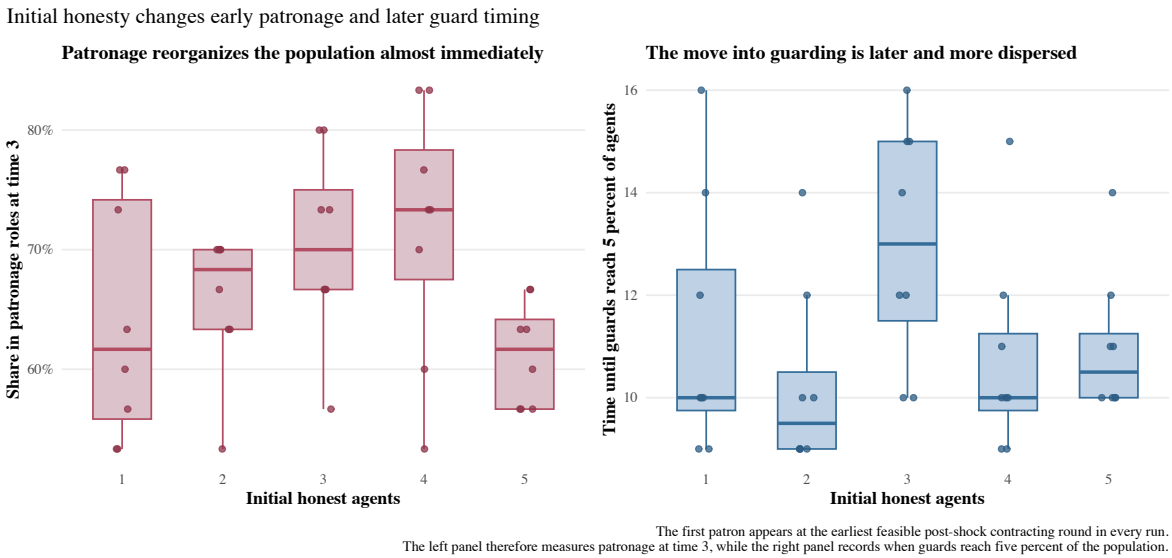

Figure 9: Early patronage and later guard timing. The first patron appears at the earliest feasible post-shock contracting round in every run, so the left panel reports the share of patrons and hired thieves at time 3. The right panel reports the time at which guards reach five percent of the population. The figure demonstrates immediate organizational change after the honest shock and a later, more dispersed movement into enforcement.

the scale of the patronage complex at time 3, after the first hiring cascade has unfolded. The right panel turns to the later margin that the chronology does not pin down mechanically and records the date at which guards reach five percent of the population. Broader honest shocks do not produce a smooth one-for-one acceleration in this clustered-shock sweep, because enlarging the initial honest set changes the local geometry of the broken cycle as well as its size. What the figure does show clearly is immediate reorganization into patronage and a later, more dispersed transition into enforcement once wealth concentration and prey depletion have progressed far enough.

Coexistence reveals institutional layering and rules out a synchronized regime switch. In those paths, some elites have crossed the guard threshold while others still rely on patronage. The resulting society is neither one of decentralized theft nor one of uniform protection. It is a mixed order in which different technologies of control are used at the same time by households in different parts of the wealth distribution. That layering matters substantively. The move from patronage to guarding is itself unequal and sequential. Some elites buy security before others can, which stabilizes their lead.

Table 3 compresses the same sequence across the baseline perturbation, patronage, and enforcement simulations. It summarizes successive layers of one model after the timing, regime, and occupa-

Table 3: Simulation summary. The first honest perturbation produces a local distributional break, patronage transforms that break into durable stratification, and enforcement preserves or deepens concentrated wealth instead of reversing it.

Scenario	Gini	Top decile	First patron	First guard	Regime
Baseline perturbation	0.231	0.200	Not reached	Not reached	Transition
Patronage dynamics	0.575	0.296	2	Not reached	Transition
Enforcement dynamics	0.640	0.306	2	7	Guard regime

tional evidence has been laid out. The baseline perturbation is the least unequal of the three simulated environments. Patronage produces a larger rise in concentration because it adds a contract technology that channels loot upward. Enforcement then preserves and slightly intensifies that concentration rather than reversing it. In that sense the table mirrors Proposition 3.12. Distributional divergence is already visible once reciprocity breaks. Later regimes magnify that divergence or lock it in through new organizational choices.

5.5 Network reorganization and welfare

Figure 10 makes the organizational change visible in network form. On the first honest night the graph is still dominated by direct theft edges arranged around the original cycle. In the middle snapshot, contract links connect patrons to hired thieves and theft edges point from those thieves toward vulnerable poor households. In the later snapshot, guard links radiate from elite households while direct theft edges thin out. More than wealth is being redistributed. The pattern of social linkage itself changes. Horizontal predation gives way to vertical contracting, which later gives way to selective protection.

The welfare consequences appear in Figure 11. Gross wealth and utilitarian welfare do not move together. Aggregate wealth declines steadily because theft, punishment, and protection burn resources even when much of the interaction is redistributive in form. Welfare deteriorates more sharply because redistribution toward the top lowers the concave utility sum and because predation, contracting, punishment, and guarding all consume resources. This is the logic of Proposition 3.12. Low welfare in the transition does not arise from honesty itself. It arises from the unequal and resource-consuming adjustment that follows an incomplete honest shock.

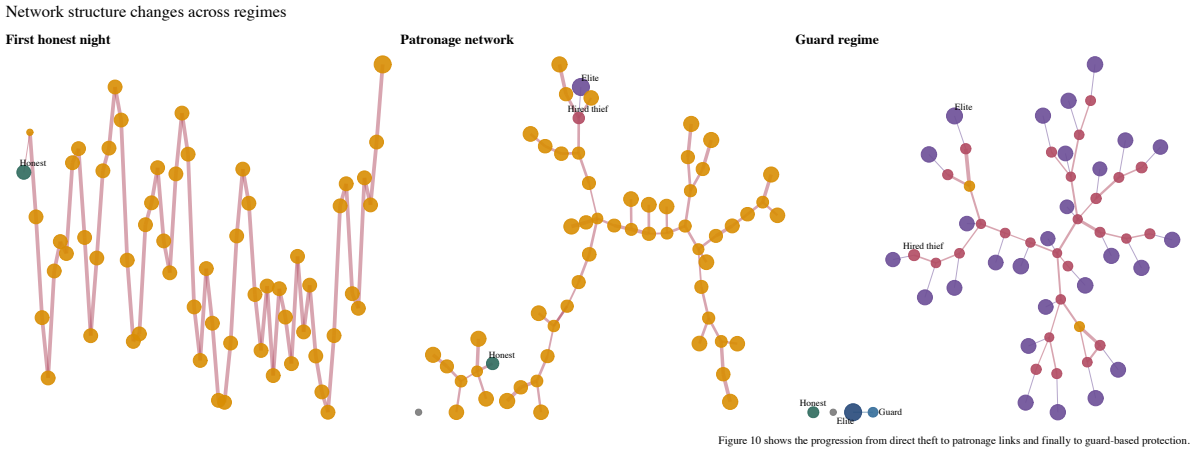

Figure 10: Network snapshots across regimes. Direct theft edges dominate at first, contract edges later connect patrons to thieves, and guard edges eventually organize elite protection. The figure demonstrates the changing network structure of predation and control.

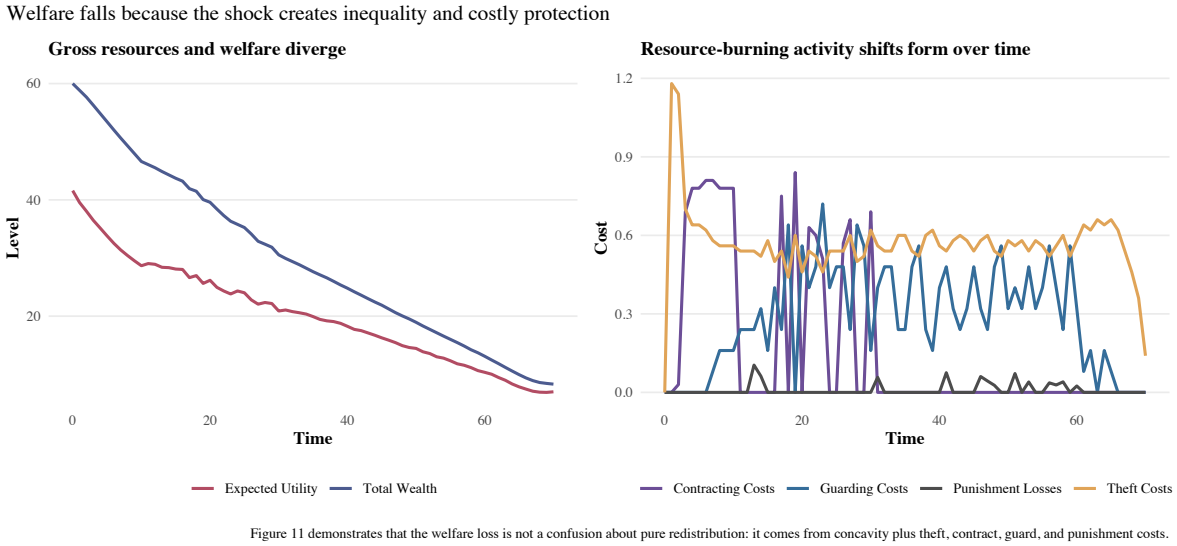

Figure 11: Welfare decomposition. Aggregate wealth and utilitarian welfare diverge because the transition adds inequality and resource burning activity in the form of theft, contracts, punishment, and guarding. The figure demonstrates the welfare mechanism directly.

The figure follows one baseline enforcement path rather than a comparison across parameter regions, so its evidentiary role is narrower than the corollary. What it shows directly is a shift in the composition of resource use. Theft costs dominate early, contracting costs spike during patronage, and guarding costs become persistent later on. Guard adoption therefore changes how resources are burned without restoring the earlier welfare level. The protected order that emerges at the end still rests on concentrated wealth accumulated during the predatory transition.

5.6 Robustness

Figure 12 collects the main one-dimensional robustness exercises in compact form. Final welfare is most responsive to theft intensity and occupied-house vulnerability in this sweep, while final inequality moves within a narrower band. The point markers also show how little regime classification moves once the system is already in the region where guarding is attractive. Among the four margins shown, only the theft-intensity sweep changes the dominant regime, passing from transition to guard and then to coexistence as extraction becomes harsher. The other sweeps mainly alter the severity of the outcome rather than the identity of the final regime. These panels therefore support a restrained robustness claim: the mechanism does not depend on a single hand-picked calibration, but some margins move welfare far more than they move regime labels.

These failure cases are informative because the mechanism operates only under a distinct and interpretable set of conditions. High contracting costs and weak targeting advantage compress the delegation surplus, while low occupied-house vulnerability reduces the value of moving from presence alone to organized protection. The sweeps therefore identify which margins attenuate the transition without pretending that every one-dimensional variation redraws the institutional map.

Figure 7 remains the more informative display for collapse. There the collapse cells are neither the moral opposite of balanced reciprocity nor a sign of numerical instability. They identify paths in which predation and protection together destroy the social base that made them profitable. Poor households become too depleted to sustain patronage, and inequality becomes so sharp that a small protected core cannot reorganize the rest into a stable order. This possibility matters because it prevents the model

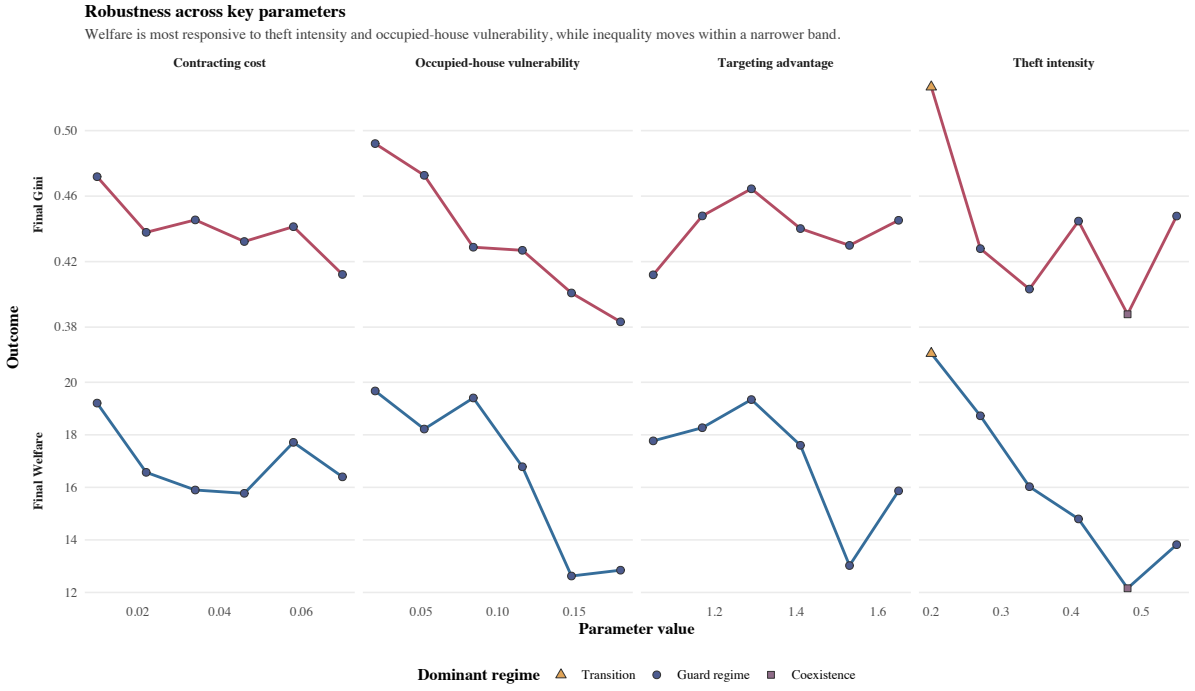

Figure 12: Robustness panels. The top row reports final inequality, the bottom row reports final welfare, and point markers identify the dominant regime reached at each parameter value. The figure demonstrates that the mechanism does not depend on a single hand-picked calibration, while also showing that some margins move welfare more than regime labels.

from telling a falsely deterministic story in which every corrupt equilibrium matures smoothly into guarded hierarchy.

6 Discussion and Conclusion

The sequence developed here isolates a form of order that remains stable while saturated with predation. Equality survives there only in the thin accounting sense that every loss is offset by a corresponding gain, and once that accounting is disturbed the successor need not be cleaner, more egalitarian, or morally mixed in any interesting way. It can instead become a harder arrangement in which the distribution created during the transition is defended more effectively than it was created. The stages of that movement remain distinct: balanced reciprocity governs the unit benchmark, a permanent honest perturbation destroys stationarity once exposure becomes wealth dependent, delegated predation emerges when wealth and vulnerability diverge enough to create surplus from hiring, and organized protection arrives when concentrated wealth becomes worth defending and the prey base has already been thinned. Welfare follows the same sequence. Honesty does not become socially harmful; welfare falls because a local rupture in a corrupt equilibrium is followed by unequal adjustment when no broader institutional change accompanies it. Some agents reduce exposure by staying home, then by hiring thieves, and later by hiring guards, while others become more exposed as their outside options weaken with repeated loss. Concavity in wealth, theft effort costs, contracting costs, guarding costs, and punishment losses all move in the same direction.

Guards and police appear only after concentrated wealth has become worth protecting and labor can be shifted from predation to protection without altering who controls the terms of social order. The move to guarding therefore replaces one order with another; it does not mark the first appearance of order. Simulation is useful precisely because that replacement unfolds through state dependence that is hard to characterize in closed form once many heterogeneous agents interact: exact event time makes the first honest night visible, patronage and enforcement paths show how the payoff comparisons behave after symmetry has broken, and the phase diagram together with the robustness panels

show where the mechanism survives and where it gives way to coexistence or collapse. Production is absent, so the analysis concerns the redistribution, dissipation, and protection of wealth rather than its endogenous creation, and the final regime remains protection of accumulated spoils rather than a full productive property-rights order. The case studied is also the one that gives the fable its force, namely a small honest rupture inside a society whose predatory balance relied on universal participation, while a broader coalition of honest agents could generate a different transition.

The paradox that made the story worth modeling remains: universal theft can look balanced, honesty can break that balance without redeeming the society, and the calmer order that follows can also be the more unequal order. Those claims now rest on explicit conditions. A balanced theft cycle can persist under symmetry, a permanent honest perturbation destroys stationarity once exposure becomes wealth dependent, patronage requires surplus over poor outside options, and guarding becomes privately optimal once concentrated wealth and prey depletion change the ranking of control technologies. The computational evidence completes the sequence by showing a precise first night asymmetry, a durable widening of the distribution, a contract regime grounded in explicit surplus, and a later move toward protection that stabilizes gains accumulated during the predatory transition. Calvino remains central because the story locates the whole movement with unusual economy.

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Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data Availability

All code, generated simulation outputs, and build scripts required to reproduce the figures and tables are available in the project repository at github.com/cheikh-a/black_sheep_geb. A versioned archival release can be deposited at the time of publication if the journal requests a persistent repository record.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

Generative AI and AI-assisted tools were used under author supervision during revision of the manuscript to support language editing, code debugging, figure-production scripting, and consistency checks. All theoretical claims, modeling choices, simulations, interpretations, and editorial decisions were reviewed and approved by the author, who takes full responsibility for the content of the article.

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A Appendix

A.1 Proofs

This appendix collects the proof details omitted from the main text. None of the omitted steps is especially difficult. The purpose of placing them here is simply to keep the main text readable while making the logical dependence of the results explicit. Throughout, the benchmark unit loot is $\tau = 1$ unless otherwise noted.

Proof of Lemma 3.2

Fix a balanced theft cycle σ and suppose every agent enters the period with wealth $\bar{w}$. By definition of a balanced cycle, each agent i undertakes exactly one successful outgoing raid against $\sigma(i)$ and suffers exactly one incoming successful raid by $\sigma^{-1}(i)$. Equation (3.1) therefore becomes

$$w_{i,t+1} = \bar{w} - \tau + \tau = \bar{w}$$

for every agent i. Summing across agents yields

$$\sum_{i=1}^{N} w_{i,t+1} = \sum_{i=1}^{N} \bar{w} = N\bar{w} = \sum_{i=1}^{N} w_{it}.$$

Hence wealth is preserved both agent by agent and in aggregate. □

Proof of Proposition 3.3

Take an arbitrary agent i in a balanced theft cycle. If i follows the prescribed action, he pays the effort cost c_T, loses one unit to the incoming raid, and gains one unit from his outgoing raid, finishing with $\bar{w} - c_T$. Consider three classes of unilateral deviations.

First, suppose i deviates by raiding a different victim while still going out to steal. Since all non-deviating households remain absent under the balanced-cycle profile, the alternative victim is still vulnerable. The deviator still obtains one unit of loot and still suffers the incoming raid from $\sigma^{-1}(i)$, so the deviating payoff remains $\bar{w} - c_T$. Such a deviation is therefore payoff equivalent.

Second, suppose i deviates by staying home. Then the raid on i fails because the house is occupied, but i gives up the one unit he would otherwise have taken from $\sigma(i)$. The deviating payoff is $\bar{w} - \tau$. Since $c_T \leq \tau$, following the cycle yields $\bar{w} - c_T \geq \bar{w} - \tau$, so staying home is not strictly profitable.

Third, suppose i contemplates a mixed deviation over alternative victims and staying home. Since neither pure deviation is strictly profitable, no mixture of them is strictly profitable either. Hence no unilateral deviation improves on the balanced-cycle payoff, and the profile is a weak Nash equilibrium.

□

Proof of Lemma 3.4

Let h be the permanently honest agent, $p = \sigma^{-1}(h)$ his predecessor in the original cycle, and $q = \sigma(h)$ his successor. At $t = 0$, agent h stays home. Because occupied houses cannot be burgled in the benchmark environment, the raid by p against h fails, so p gains no loot from his planned target. Agent h also undertakes no outgoing raid, so q is not robbed by h. All other agents follow the original cycle exactly.

Now compute end-of-night wealth. Agent h neither loses nor gains, so $w_{h1} = \bar{w}$. Agent p is still robbed by his own predecessor but gains nothing from his failed attempt on h, so $w_{p1} = \bar{w} - \tau$. Agent q still robs his own target but is no longer robbed by h, so $w_{q1} = \bar{w} + \tau$. Every other agent both loses one unit and gains one unit, hence remains at $\bar{w}$. This proves (3.2). □

Proof of Theorem 3.6

By Lemma 3.4, the successor q enters period $t = 1$ with wealth $w_{q1} = \bar{w} + \tau$. The expected one period net payoff from personal theft at wealth w is

$$\pi^S(w) = \tau - \rho w - c_T.$$

The payoff from staying home is normalized to zero. Therefore the agent strictly prefers staying home whenever $w > \widehat{w} = (\tau - c_T)/\rho$.

Assumption 3.5 states that $\bar{w} < \widehat{w} \leq \bar{w} + \tau$. The lower inequality ensures that at the symmetric benchmark wealth level $\bar{w}$ an agent weakly prefers theft or is indifferent, which is why the original cycle can be sustained before the honest perturbation. The upper inequality ensures that after the first night windfall the successor's wealth lies at or above the cutoff. Hence

$$w_{q1} = \bar{w} + \tau \geq \widehat{w}$$

and, if the inequality is strict, $\pi^S(w_{q1}) < 0$; if equality holds, the agent is indifferent and any stationary selection that retains the original cycle is knife-edge. In either case, once the honest perturbation moves the successor to or beyond the threshold, the original balanced pattern is no longer a robust stationary continuation for all agents other than h.

To prove the theorem as stated, suppose by contradiction that there exists a stationary Markov equilibrium preserving the original balanced theft pattern among all nonhonest agents. Then q must choose personal theft at state w_{q1}. But the threshold condition implies that staying home weakly dominates, and under strict inequality strictly dominates, personal theft at exactly that state. This contradicts the

alleged stationary continuation. Therefore no such stationary Markov equilibrium exists. □

Proof of Lemma 3.7

Fix a rich patron i and a candidate poor thief j. Let delegated gross loot be L_t, contracting cost k, and the outside option of the poor worker o_{jt}. A contract with terms (s_{ijt}, α_{ijt}) is feasible if the participation constraint of the thief holds:

$$s_{ijt} + \alpha_{ijt} L_t \geq o_{jt}.$$

The net return of the patron from the contract is

$$(1 - \alpha_{ijt})L_t - s_{ijt} - k = L_t - (s_{ijt} + \alpha_{ijt} L_t) - k.$$

For some contract to be mutually acceptable, there must exist a transfer to the thief that is at least o_{jt} and leaves the patron weakly nonnegative. This requires

$$o_{jt} \leq s_{ijt} + \alpha_{ijt} L_t \leq L_t - k.$$

Such an interval exists if and only if $L_t - k - o_{jt} \geq 0$, with strict inequality for strictly positive joint surplus. This is exactly condition (3.12). □

Proof of Proposition 3.8

Suppose (3.12) holds, so total joint surplus is

$$\Sigma_t = L_t - k - o_{jt} > 0.$$

Under the bargaining rule in (3.11), the thief receives

$$o_{jt} + (1 - \beta)\Sigma_t$$

in expected value, while the patron receives the residual

$$L_t - k - [o_{jt} + (1-\beta)\Sigma_t] = L_t - k - o_{jt} - (1-\beta)\Sigma_t = \beta\Sigma_t.$$

This proves (3.13). Hiring dominates abstention whenever $\beta\Sigma_t > 0$, and it dominates personal theft whenever

$$\beta\Sigma_t > m_t - \rho w_{it} - c_T.$$

Combining the two comparisons yields (3.14). □

Proof of Corollary 3.9

Using (3.13) together with $L_t = \phi_t \gamma \bar{w}_t^H$ and $o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T$, patron payoff can be written as

$$\pi_{ijt}^{\text{hire}} = \beta \left(\phi_t \gamma \bar{w}_t^H - k - \lambda_t \gamma \bar{w}_t^P + c_T \right).$$

Differentiating yields

$$\frac{\partial \pi_{ijt}^{\text{hire}}}{\partial \phi_t} = \beta\gamma\bar{w}_t^H > 0, \qquad \frac{\partial \pi_{ijt}^{\text{hire}}}{\partial \lambda_t} = -\beta\gamma\bar{w}_t^P < 0, \qquad \frac{\partial \pi_{ijt}^{\text{hire}}}{\partial k} = -\beta < 0.$$

Thus patron payoff rises with delegation quality and falls with poor outside options and contracting cost. Since patronage is chosen when (3.14) holds, any change that raises the left-hand side of that inequality, holding the right-hand side fixed, expands the set of states in which patronage dominates. □

Proof of Theorem 3.10

Elite agent i compares the payoff from guarding,

$$\pi_{it}^{\text{guard}} = q_t \chi_t \gamma_E w_{it} - g,$$

with the payoff from one more patron-thief contract,

$$\pi_{ijt}^{\text{hire}} = \beta \left(L_t - k - o_{jt} \right).$$

Guarding is optimal if and only if

$$q_t \chi_t \gamma_E w_{it} - g \geq \beta \left(L_t - k - o_{jt} \right),$$

which rearranges to

$$w_{it} \geq \frac{\beta \left(L_t - k - o_{jt} \right) + g}{q_t \chi_t \gamma_E} \equiv \widehat{w}_t^G.$$

This proves the threshold expression in (3.18). To show that the threshold falls as the poor prey base is depleted, note that both $L_t = \phi_t \gamma \bar{w}_t^H$ and $o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T$ depend positively on the wealth of the poor. Under the maintained assumption that delegated targeting advantage shrinks more slowly than the broad poor outside option, depletion lowers net delegated surplus $L_t - o_{jt}$ and hence lowers the threshold wealth level at which guarding dominates. □

Proof of Corollary 3.11

Write the guarding threshold as

$$\widehat{w}_t^G = \frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t \gamma_E}.$$

Direct differentiation gives

$$\frac{\partial \widehat{w}_t^G}{\partial q_t} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t^2 \chi_t \gamma_E} < 0, \qquad \frac{\partial \widehat{w}_t^G}{\partial \chi_t} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t^2 \gamma_E} < 0,$$

$$\frac{\partial \widehat{w}_t^G}{\partial \gamma_E} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t \gamma_E^2} < 0, \qquad \frac{\partial \widehat{w}_t^G}{\partial g} = \frac{1}{q_t \chi_t \gamma_E} > 0.$$

Finally,

$$\frac{\partial \widehat{w}_t^G}{\partial\{\beta(L_t - k - o_{jt})\}} = \frac{1}{q_t \chi_t \gamma_E} > 0.$$

Hence stronger guarding technology or greater elite vulnerability lowers the wealth threshold for protection, while more profitable patronage or more expensive guard labor raises it. □

Proof of Proposition 3.12

Fix date t and suppose aggregate wealth $\sum_i w_{it}$ is fixed. Let u be concave. If one wealth vector is a mean preserving spread of another, then by the Hardy-Littlewood-Polya majorization theorem, or equivalently by Jensen's inequality applied to pairwise spreads, the more unequal vector yields a weakly smaller value of $\sum_i u(w_{it})$. This proves the first claim.

Now compare two regimes with the same aggregate wealth. If one regime has weakly greater inequality, then its utility sum is weakly lower. If, in addition, at least one of the cost terms n_t^S, n_t^H, n_t^G, or Ψ_t is strictly larger and none is smaller, then subtracting those costs from the already weakly smaller utility sum yields strictly lower period welfare $\mathcal{W}_t$. This proves the second claim. □

Proof of Corollary 3.13

Fix the wealth distribution. The difference between period welfare under guarding and period welfare under patronage is

$$\Delta\mathcal{W}_t = -gn_t^G - c_T n_t^{S,\text{guard}} - kn_t^{H,\text{guard}} - \Psi_t^{\text{guard}} + c_T n_t^{S,\text{pat}} + kn_t^{H,\text{pat}} + \Psi_t^{\text{pat}}.$$

If the reduction in theft effort cost, contracting cost, and punishment loss exceeds the new guard wage bill, then $\Delta\mathcal{W}_t > 0$. Since the wealth distribution is fixed by hypothesis, the utility term cancels. Hence guarding raises welfare relative to patronage under the stated condition. □

A.2 Alternative assumptions and scope

Several assumptions deserve comment because they determine both what the paper proves and what it leaves open. The first is the use of a unit transfer benchmark followed by a proportional transfer extension. A single proportional model could in principle have been used throughout. The difficulty is that the egalitarian balance at the beginning of the story by Calvino is most transparent when every successful raid transfers the same unit amount and every agent has exactly one incoming and one outgoing link. The benchmark therefore isolates the cycle cleanly. Once the paper turns to inequality and class formation, however, the unit transfer environment becomes too rigid because it cannot represent the idea that richer agents risk more by leaving home. The proportional extension is introduced at exactly that point and no earlier.

The second assumption concerns stationary Markov strategies in the honest-perturbation theorem. A reader might ask whether history-dependent punishments could sustain a one-honest continuation even after the first windfall. The answer may well be yes in some environments, but that is not the object of the theorem. The theorem isolates the local best-response logic that the simulation later implements. It says that once the wealth of the successor moves above the exposure threshold, the original cycle is not a stationary continuation. That is the appropriate statement for a paper interested in regime transition and not in the folk-theorem support of arbitrary histories.

The third assumption concerns the outside option of poor workers. The paper models it as weak because poor agents compete for a depleted set of vulnerable targets and because bargaining power is tilted toward patrons. One could instead model search frictions, repeated matching, or explicit auctioning of predatory labor. Any of those would add realism, but the key analytical requirement would remain the same: patronage must create a wedge between delegated gross returns and the best alternative available to poor workers. Without that wedge, the contract market has no surplus to divide.

The fourth assumption concerns guarding. In the model, guarding becomes attractive once occupancy alone no longer fully protects concentrated wealth after patronage weakens and poaching risk rises. This is summarized by the residual elite-risk term χ_t. A more elaborate microfoundation could model betrayal by former hired thieves, explicit coordination among freelancers, or the public-good as-

pect of policing. The reduced form used here is meant only to capture the fact that highly concentrated wealth becomes worth defending with a more dedicated technology than merely staying home.

A.3 Simulation implementation details

This subsection records the simulation algorithm in enough detail that a reader can map every figure in the paper to a precise computational step. The code itself is the authoritative implementation, but the written algorithm helps align the appendix with the manuscript's notation.

1. Initialize N agents with common wealth $\bar{w}$, assign the balanced theft permutation σ, and designate the initial honest set. In the baseline path the honest set contains one permanent honest agent.

2. Run the deterministic benchmark perturbation. On the first night the honest agent stays home, all other agents follow the original cycle, and wealth is updated according to the accounting in Lemma 3.4.

3. In later benchmark periods, let each nonhonest agent compare current wealth to the threshold $\widehat{w}$ in (3.4). Agents above the threshold stay home; agents below it continue the cycle.

4. For the proportional transfer simulations, reinitialize from the common wealth state and apply the first honest perturbation, then classify rich and elite agents each period using the threshold multipliers θ_R and θ_E.

5. Compute candidate personal theft returns for rich agents using (3.6). Compute candidate contract surplus using (3.10) and accept only contracts satisfying (3.12). Order patrons from richest to poorest and match them to the poorest available workers.

6. Assign hired thieves to the high value poor target set H_t, freelancers to vulnerable households ranked by expected loot, and guards to elite households satisfying (3.18). Apply the guard acceptance condition (3.16).

7. Realize raids sequentially. Each realized transfer is based on the victim's remaining wealth at the moment of attack, so multiple attacks on the same house have diminishing returns within a period.

8. Apply wages, loot shares, contracting costs, guard wages, and punishment losses. Update the wealth vector and the role vector.

9. Compute inequality statistics, welfare components, first entry times into patronage and guarding, and the regime label for the period. Store these outputs before advancing to the next date.

10. Repeat until the simulation horizon is reached. Store processed data, figures, and tables through the reproducible build pipeline documented in the replication materials.

Three implementation choices are especially important. First, wealth changes are recorded in realized rather than merely expected values, but the occupational comparisons are based on expected payoff formulas that correspond to the theory. Second, occupied but unguarded houses are treated differently in the first night benchmark and the later stochastic simulations. In the benchmark they are perfectly safe because that is what the instability theorem requires. In the later simulations they are imperfectly safe because concentrated wealth eventually requires a dedicated protection technology. Third, the code distinguishes clearly between rich households hiring poor thieves to rob poor targets and elites later hiring poor guards to defend elite assets. This distinction was blurred in the earlier draft and is now maintained throughout.

A.4 Supplementary robustness and regime frequencies

The phase diagram in the main text summarizes the most informative parameter plane, but the code evaluates broader robustness variations. These include changes in the outside option parameter, the number of initially honest agents, the guard punishment probability, and the vulnerability of occupied but unguarded homes. The overall message is the same as in the main text: the mechanism is robust within a substantial region, but it does not survive when the analytical conditions behind the main propositions are switched off. If delegated theft has no surplus, patronage disappears. If guarding is ineffective or too expensive, elites continue with patronage or the system collapses. If theft intensity is too low, the society can remain close to the transition regime for long periods.

Table 4: Dominant regime frequencies in the main parameter sweep. The table reports how often each regime is the dominant late period outcome over the grid used to produce the phase diagram.

Dominant regime	Frequency
Guard regime	18
Coexistence	6
Transition	5
Collapse	3

Table 4 reports the frequency with which each dominant regime appears in the main parameter sweep. These frequencies are not structural predictions, because they depend on the chosen grid. Their value is descriptive. They show that transition, coexistence, guarding, and collapse all appear in the sweep, which confirms that the model generates multiple successor regimes and not a single canned trajectory.

A.5 Annotated passage map for *The Black Sheep*

The translated story is not reproduced in full here because the text remains copyrighted. What can be included usefully is a passage map that identifies the parts of the story on which the model relies and the analytical object attached to each part. Figure 13 does that in paraphrased form.[2]

[2]The map is based on the sequence in Calvino (1993). It is intended as an interpretive guide to the model, not as a substitute for the story.

Story movement 1. Everyone steals at night and everyone is robbed in turn. Equality survives because outgoing predation and incoming predation offset one another.

Model fit. Section 3 turns this opening into the unit transfer benchmark and the balanced theft cycle of Definition 3.1 and Proposition 3.3.

Story movement 2. One man remains at home instead of stealing. The immediate effect is local, not universal. A planned burglar returns empty handed, and another household keeps what it otherwise would have lost.

Model fit. Lemma 3.4 records the exact first night accounting. Theorem 3.6 shows why that local windfall destroys stationarity once exposure rises with wealth.

Story movement 3. Inequality appears. Some households become rich because the broken chain no longer offsets gains and losses symmetrically.

Model fit. The proportional theft extension introduces wealth dependent exposure and turns class formation into a dynamic consequence, not a narrative assertion.

Story movement 4. The rich no longer steal personally. They hire the poor to steal for them.

Model fit. Lemma 3.7 and Proposition 3.8 derive the feasibility and surplus conditions for patronage. Figure 6 studies how that surplus moves with targeting advantage and outside options.

Story movement 5. Later the rich hire guards and police. Violence becomes more selective, but the unequal order remains.

Model fit. Theorem 3.10 gives the guarding threshold. The welfare analysis and Figures 7 through 11 show how organized protection can stabilize the spoils of the earlier transition.

Figure 13: Annotated passage map of the story by Calvino. The figure paraphrases the key narrative movements and indicates the point in the paper where each movement is formalized or simulated.